

JASC DATA PROCESSOR

Comprehensive User Manual

A lightweight, modular desktop application for data ingestion, visualization, mathematical transforms, statistics, and curve fitting.

Reference Manual & Guide

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1. Installation and First Launch

System requirements

- Python 3.10 or later.
- A desktop environment capable of displaying a Qt application.
- The packages in requirements.txt, including PySide6, pandas, Polars, NumPy, Matplotlib, and SciPy.

Install dependencies

Open a terminal in the repository root and run:

```
python -m pip install -r requirements.txt
```

Using a virtual environment is recommended so that JASC's package versions remain isolated from other Python projects.

Start JASC

Run the application from the repository root:

```
python main.py
```

main.py is the supported entry point. Do not run ui/main_window.py directly. In the supplied VS Code workspace, the Run JASC Data Processor launch configuration starts the correct entry point.

At first launch, JASC opens an empty workspace. Create a sheet, import data, or open a .jasc project to begin.

File types at a glance

Purpose	Supported format
Tabular import	.csv, .txt, .xlsx, .xlsm (Excel worksheet values only)
Tabular export	.csv, .txt / tab-delimited text
Project save/open	.jasc
Recorded macro save/load	.jasc-macro.json (separate from project files)
Python API recording	.py script containing run(session)
Figure digitizer	PNG, JPEG, BMP, TIFF, GIF, and other image formats readable by Qt

Plot image export	Formats offered by the embedded Matplotlib toolbar

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2. Application Window, Menus, and Shortcuts

Main layout

```
+-------------------------------------------------------------------------+
| File | Edit | View | Data | Analysis | Plot                            |
+-------------------------------------------------------------------------+
| Main toolbar: project controls, new sheet, import, digitizer, export    |
+-------------------+---------------------------------+-------------------+
| Workspace         | Central work area               | Plot Inspector    |
|                   |                                 |                   |
| Datasets          | Data tab: worksheet editor      | Series            |
| Saved Plots       | Plots: independent plot tabs    | Axes              |
|                   | and navigation toolbar          | Legend            |
|                   |                                 | Locator           |
+-------------------+---------------------------------+-------------------+
| Status bar: active operation, warnings, progress, and result messages   |
+-------------------------------------------------------------------------+
```

- Workspace Navigator: The left dock lists datasets and project plots. Select a dataset to display it in Data; select a plot to open or activate its independent tab without replacing another plot.
- Data tab: Shows the selected worksheet and a summary containing its dimensions, memory estimate, metadata/formula counts, and backend.
- Plots tab: Contains one tab per open plot, each with its own canvas, Matplotlib navigation toolbar, editor, and view state. Use + for another plot and x to close a view while keeping its plot in the project.
- Plot Inspector: The right dock is synchronized with the Plot tab and contains Series, Axes, Legend, and Locator controls. It is hidden while the Data tab is active.
- Status bar: Reports completed operations, selected outputs, large-data warnings, and batch progress summaries.

JASC automatically follows the operating-system light or dark palette and reapplies theme-aware table, input, dock, and dialog styling when the Qt palette changes.

Checkboxes, radios, and toggle buttons now have explicit visible states, including partial/disabled states. Inputs have contrasting boundaries and keyboard focus outlines; navigation icons adapt to checked and highlighted backgrounds. Colour swatches choose readable text. Comparison readouts use their own contrasting surface; the scientific figure/export palette remains controlled by plot settings.

Window dimensions are calculated in Qt logical pixels from the active monitor's available work area, so operating-system scaling and taskbar space are respected. Resizing or moving the window between monitors triggers a debounced dock rebalance. Inspector forms wrap long rows and keep their editors usable; Plot actions that do not fit move into the toolbar overflow menu. In a very narrow Plot layout, JASC temporarily collapses the Workspace Navigator to protect the plot canvas. Selecting View > Show Workspace Navigator explicitly restores it even at that width.

View controls

The View menu can independently show or hide:

- Workspace Navigator: the complete left dock.
- Workspace Datasheets: the dataset section within the navigator.
- Saved Plots: the saved-plot section within the navigator.
- Plot Inspector: the right plot-control dock.

Use View > Reset View to Default to restore the normal dock arrangement and visibility.

Mouse and context actions

JASC follows the desktop convention used by spreadsheet and scientific-analysis software:

- Left-click selects a worksheet, saved plot, cell, queued series, or locator. Double-click a worksheet cell to edit it, a saved plot to load it, or a rendered plot element to open its focused editor.
- Right-click opens the actions relevant to the item under the pointer. Worksheet menus are available from cells, empty table space, row headers, and column headers. Right-clicking an unselected worksheet cell, row, column, dataset, series, locator, or saved plot makes that item the context target first.

The dataset navigator menu includes Open Sheet, Create Plot from Dataset, export, Data Preparation and Analysis submenus, duplication, rename, and removal. Create Plot from Dataset starts a protected empty draft and prefills its dataset/column mappings; choose the subtype and click Add Series to render it. Worksheet menus include value, row, and column copying; normal and transposed paste; cell clearing; worksheet resizing; Row and Column submenus; plotting; Data Preparation; and Analysis. Saved-plot, queued-series, and locator lists expose their corresponding create/save/load/update/revert/duplicate/remove actions. Disabled actions remain visible when the current selection or state does not support them.

Menu reference

Menu	Commands
File	Open Project, Save Project, four explicit Save As security modes, Signing Identities, New Blank Sheet, Close Project, Import Data, Import From Figure, Export Dataset, Exit

Edit	Undo, Redo, Clear Selected Cell Values, Delete Selected Rows, Delete Selected Columns, Paste Values Transposed
View	Navigator/section/inspector visibility and Reset View to Default
Data	Duplicate/Rename/Remove Dataset; Add/Insert Row or Column; Resize Sheet; Fill Range; Smart Fill Selection; all Data Preparation tools; Column Metadata; Apply Formula
Analysis	Derivative, Integral, Smoothing, Interpolation, FFT, Inverse FFT, Statistics, Baseline Correction, Peak Analysis, Raman Analysis, Custom Fit Models, Curve Fit
Macros	Record Macro, Stop Recording, View Recorded API Calls, Save Python API Script, Review / Run Batch Recipe, Load Batch Recipe, Save Batch Recipe
Plot	New Plot, Rename Current Plot, Copy Current Plot to Clipboard, Open/Duplicate/Delete Selected Plot

Keyboard shortcuts

Shortcut	Action
Ctrl+O	Open project
Ctrl+S	Save project
Ctrl+Shift+S	Save project as
Ctrl+W	Close project
Ctrl+N	New blank sheet
Ctrl+I	Import data
Ctrl+E	Export selected dataset
Ctrl+Z	Undo worksheet edit
Ctrl+Y	Redo worksheet edit
Delete	Clear values in selected worksheet cells; rows and columns remain
Shift+Delete	Delete selected rows after confirmation
Ctrl+Delete	Delete selected columns after confirmation
Ctrl+C	Copy selected worksheet values
Ctrl+V	Paste worksheet values at the active cell
Ctrl+Shift+V	Paste the clipboard matrix transposed at the active cell
Ctrl+Shift+F	Apply Formula
Ctrl+Shift+C	Curve Fit
Ctrl+Shift+R	Reset view layout
Ctrl+Alt+N	New empty plot

Ctrl+Alt+S	Rename current plot
Ctrl+Alt+C	Copy the current live plot visualization to the clipboard

---

3. Projects, Datasets, and the Workspace

Workspace concepts

- A dataset is a named pandas worksheet with a table, formulas, column metadata, source path, and backend label.
- A plot document is an independent named project object containing series, axes, legend, locators, style, figure settings, and pending editor/view state. Its identity stays stable when renamed.
- A custom fit model is an expression-based fitting function stored with the project.
- The workspace is the collection of all three.

Names are made unique automatically. If Sheet already exists, another dataset requested with that name becomes Sheet (2), then Sheet (3), and so on. Saved plots and custom models use equivalent unique-name behavior.

Open, save, and close a project

1. Choose File > Open Project... to load an open .jasc or encrypted .jascx file. Encrypted projects prompt for their password. Opening a project replaces the current workspace after confirmation.
2. Choose File > Save Project to save to the current project path. The first save opens the Save As dialog.
3. Choose File > Save Open Project As... for a normal .jasc, or File > Save Encrypted Project As... for password-protected .jascx. Encrypted saves ask for Encryption password and Confirm encryption password; the password is retained only for the current open application session so later saves do not repeatedly prompt.
4. Choose File > Save Signed Open Project As... for signed .jasc, or File > Save Signed + Encrypted Project As... for signed .jascx. These are separate menu actions rather than hidden file-type choices.
5. Choose File > Close Project to clear the workspace without closing JASC. This action warns that unsaved changes will be lost.

Save before using File > Exit or closing the application window. Project-level automatic save and recovery are not implemented.

What a `.jasc` project preserves

A project is a ZIP archive containing CSV worksheet payloads and JSON metadata. It preserves:

- Dataset names, values, source paths, and backend labels.
- Worksheet-level source/import metadata, parsed instrument fields, and exact raw text preambles.
- Column formulas.

- Column roles, value types, precision, units, long names, and comments.
- Operation-provenance comments written by JASC.
- Resumable figure-extraction sessions attached to digitized datasets, including an embedded working image and X/Y calibration; colour-plot sessions also retain their empirical colour-bar calibration.
- Saved plot definitions, including plot series and appearance settings.
- Project-scoped custom fit models.
- Named spectral-analysis sessions with readable metric, fitted-curve, parameter,

and detected-peak tables plus their configuration and provenance.

Every newly saved version 8 project contains an integrity.json JASC Package Seal introduced in version 6. Its canonical manifest records every other member's exact filename, original byte size, and SHA-256 digest together with a UTC creation timestamp, random package UUID, and 256-bit random nonce. JASC sorts and deterministically serializes that manifest, combines it with the JASC-SEAL-v1 domain marker, and hashes it again to create the package seal. Each save receives a new identity and seal even when the scientific content is unchanged. Spectral-analysis internals are normal UTF-8 CSV members under analyses/, with JSON metadata in project.json, so derived results remain inspectable without JASC.

JASC verifies the seal, names, sizes, and member hashes before loading. Missing, added, renamed, duplicated, resized, or changed members, including a manifest edited without correctly recreating the complete seal, trigger a Project Integrity Warning stating that the package may have been forged or damaged. The safe default is Cancel, with an explicit Open Anyway choice for a trusted source. Version 5 hash manifests and version 4 or older projects without a manifest remain loadable for backward compatibility.

The Package Seal is designed as a practical normal-security forgery and corruption check. It makes casual CSV replacement or editing only one member hash insufficient, but it is not a digital signature or independent proof of authorship: a knowledgeable attacker who fully reproduces the documented algorithm can generate a different internally consistent package. Use encrypted .jascx when the package or threat model is sensitive.

Open and encrypted project packages

The normal .jasc format remains the recommended open-science form. It is an ordinary ZIP archive, so its CSV values and JSON metadata can be recovered without JASC. It provides transparency and portability, not secrecy or proof of authorship.

The optional .jascx form encrypts the complete .jasc package with AES-256-GCM. Its key is derived from the user password with Scrypt and a random salt; each save also uses a fresh random nonce. Authentication means that a wrong password or any ciphertext/header change is rejected before project data is loaded. Saves replace the destination atomically only after the new package has been completed. JASC never writes the password into the project.

Encryption has deliberate trade-offs:

- A .jascx package requires JASC (or a separately implemented compatible decryptor) and its password before the open ZIP/CSV contents can be recovered.
-

There is no password-recovery mechanism. Losing the password means losing access, so retain it in an appropriate password manager.

- Anyone who knows the password can decrypt, edit, and encrypt a new package. Authenticated encryption therefore protects against outsiders and accidental/unauthorized byte changes, but it does not prove which authorized person created the package.
- For publication and long-term preservation, save a separate open .jasc copy when confidentiality restrictions permit it.

Digital signatures and signing identities

JASC can optionally sign the Package Seal with Ed25519. The private key creates the signature and must remain secret; the public key verifies it and can be distributed freely. Because the signature covers the Package Seal, an attacker who edits data and completely regenerates the normal manifest still cannot create the original author's signature without the private key.

1. Choose File > Create Signing Identity..., select a .jasc-key location, enter the researcher or organization name, and protect the private key with a strong password.
2. JASC writes a password-encrypted .jasc-key private identity and a matching .jasc-pub public identity. Back up the private key securely. Share only the public file.
3. Choose File > Save Signed Open Project As... or File > Save Signed + Encrypted Project As..., then unlock the private key when the clearly labelled Private-key password prompt appears.
4. Recipients obtain the .jasc-pub file through an independent trusted channel, compare its SHA-256 fingerprint, and choose File > Trust Public Signing Key....

When a project opens, JASC reports one of four states:

- Signed and trusted: the Ed25519 signature is valid and its fingerprint is in the local trusted-signer list.
- Signature valid, signer unknown: the mathematics is valid, but the public-key fingerprint has not been independently trusted.
- Unsigned: the package has normal Package Seal protection but no author signature.
- Invalid signature: the signature is missing where required, malformed, uses a different fingerprint, or no longer matches the Package Seal. JASC treats this as an integrity failure and defaults to cancelling the open operation.

The signature record embeds the public key for portable mathematical verification, while the signed project.json declaration binds the claimed signer name and fingerprint to the sealed package. The embedded key alone does not establish identity; trust comes from comparing or importing the matching .jasc-pub fingerprint through an independent source. Signed encrypted projects are signed internally before AES-256-GCM encryption, so password authentication and author verification both apply after decryption.

Because worksheet payloads are CSV, specialized pandas dtypes are reconstructed on load rather than serialized as native binary dataframe types.

Dataset navigation and management

- Select a dataset in the navigator to display it.

- Click Duplicate Sheet, right-click a dataset and choose Duplicate Sheet, or use Data > Duplicate Dataset to create and open a fully independent copy. Values, column order and dtypes, formulas, metadata, source provenance, and backend label are retained. Copies are named Sheet Copy, Sheet Copy (2), and so on; later edits do not affect the source.
- Right-click a dataset for its common Open, Plot, Export, Data Preparation, Analysis, Duplicate, Rename, and Remove operations.
- The same management actions are available from the Data menu.
- Renaming a dataset retargets open and closed plot documents, including pending editor bindings. Deleting a dataset retains dependent plot definitions and shows missing-source feedback so bindings can be repaired.
- Deleting a dataset asks for confirmation. Dataset deletion is not part of worksheet Undo/Redo.

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4. Importing and Creating Data

Import CSV, DAT, or TXT data

Text-file inspection and reading run in the background. The status bar shows the current file and phase, byte progress while reading ordinary tables, and a Cancel button. Existing worksheets and plots remain available while files are being prepared. Cancellation discards the pending batch without adding partial worksheets; it may wait for the current parsing or conversion work unit to finish. Successfully prepared files are added together after any size warnings are resolved. If an individual file fails, the completed import identifies it in a warning and can still add the other successful files. Switching projects discards stale import results, and closing JASC waits for the worker to stop safely.

1. Choose File > Import Data... (Ctrl+I) or use the toolbar action.
2. Select one or several .csv, .dat, or .txt files. Every successfully imported file becomes an independent dataset named from its filename.
3. Review the preflight table. For each file it shows the detected text encoding, delimiter, number of leading metadata rows, header status, column count/names, and the first data values.
4. Choose a Data type. Auto (infer from file) is the default; explicit choices are General table, Single spectrum, and Spectroscopy map.
5. Choose a Spectrometer profile. Auto / generic is the default. Horiba LabSpec supports single spectra and wide spectroscopy maps; B&W Tek currently supports single spectra only. JASC stops an unsupported B&W Tek map selection instead of guessing another vendor format.
6. Leave Text encoding on Auto (detect safely) or explicitly choose UTF-8, UTF-8 with BOM, UTF-16, Windows-1252/ANSI, or Latin-1. Auto recognizes byte-order marks first, then attempts strict UTF-8, Windows-1252, and Latin-1 without silently inserting replacement characters.
7. Choose a Number format:

- Auto (infer safely) recognizes unambiguous decimal-point, decimal-comma, scientific-notation, and grouped values by column.
- Decimal point: 1,234.56 treats . as decimal and , as thousands grouping.
- Decimal comma: 1.234,56 treats , as decimal and . as thousands grouping.

8. Leave Delimiter on Auto (infer from file) or choose an explicit delimiter:

- Auto (comma, semicolon, tab, pipe, or whitespace)
- Comma
- Semicolon
- Tab
- Pipe
- Space / whitespace (one or more whitespace characters)

9. Choose Headers:

- Auto (detect for each file) is the default.
- First tabular row contains column names forces a header after any discarded metadata rows.

- No header row keeps the first tabular row as data and creates Column 1, Column 2, and so on.

10. Choose Leading metadata:

- Auto (detect and preserve leading metadata) is the default and is resolved independently for each file. These rows are excluded from the numeric table but retained with the worksheet.
- Keep all leading rows as worksheet data begins table parsing at the start of every file.
- Use a fixed number of leading metadata rows applies one explicit boundary to every selected file and retains those rows as metadata.

11. Confirm the import.

Auto examines a bounded sample before pandas reads the complete file. A file with no separator evidence imports as one column. Header, metadata, and number-format detection use structural/value-pattern heuristics, so the preflight is intentionally visible and every inference can be overridden. Values such as 12,50 and 1.234,56, or 12.50 and 1,234.56, provide clear evidence. A value such as 1.234 alone is inherently ambiguous: it may mean one point two three four or one thousand two hundred thirty-four. Auto therefore retains the backward-compatible point-decimal interpretation; select the explicit comma-decimal mode when the dot is grouping. Leading records beginning with #, $, or %, plus common key/value forms, are parsed when possible and always retained verbatim. Use the Worksheet Metadata button above the Data table or Data > Worksheet Metadata... to review source/import details, selected number format, parsed fields, and the raw preamble.

Spectrometer-oriented text support includes tab-separated decimal-comma numeric conversion used by Horiba exports, full semicolon-table import for B&W Tek single-spectrum files, and reconstruction of legacy Bruker IR text where a #c record stores the X coordinate array followed by scalar intensity values. The requested data type and spectrometer profile are retained in Worksheet Metadata.

JASC also recognizes the wide Horiba LabSpec hyperspectral-map source layout. Its first data row has two blank cells or X/Y coordinate labels followed by numeric Raman shifts; each later row is one complete spatial-pixel spectrum. During import, JASC normalizes that matrix into four numeric worksheet columns: X, Y, Raman Shift, and Intensity. Each spatial pixel therefore contributes one worksheet row per Raman shift. Pixel order and Raman-shift order, including descending axes, are preserved. Worksheet Metadata records the wide source layout, normalized worksheet layout, spatial-pixel count, shifts per pixel, unit, and axis range. Map profiles for other manufacturers remain deferred until representative formats are supplied.

Unknown records remain in the raw metadata rather than being discarded. The resolved encoding is recorded in Worksheet Metadata. If an unusual instrument encoding is not identified correctly, choose an explicit encoding and review the preflight before importing. Additional stacked/long-form/multi-acquisition vendor profiles remain planned.

When several files use Auto headers, JASC compares the results before importing. If some appear headerless or the detected names differ, choose:

- Keep Detected to import each file with its own detected structure.
- Use Generic Columns to drop detected header rows and consistently name columns Column 1, Column 2, and so on; genuinely headerless files keep their first data row.
- Cancel Batch to import nothing and revise the source selection or rules.

Unreadable files stop the preflight. Once importing begins, an individual parse failure or declined large-dataset warning is reported without removing other files imported successfully.

The imported dataset uses the source filename as its base name and remembers the source path. Files of at least 100 MiB trigger a warning before parsing. Large parsed worksheets trigger a second size/memory warning before they enter the workspace.

Import an Excel workbook

1. Choose File > Import Data... (Ctrl+I) or use the toolbar action.
2. Select a modern .xlsx or .xlsm workbook. Legacy .xls files must first be saved as .xlsx in Excel or another compatible application.
3. Review the workbook compatibility warnings. JASC inspects the workbook package for charts, pivot tables, Excel tables, images/media, chart-only tabs, macros, external links, query tables, data connections, comments, slicers, named ranges, and hidden tabs.
4. Choose Import all worksheet tabs or Choose specific worksheet tabs. Selective import is the safer default: visible worksheets start checked and hidden worksheets start unchecked. Import All includes hidden worksheets. Chart-only tabs are listed for transparency but are always excluded because they contain no worksheet grid.
5. Choose whether the first row of every selected worksheet contains column names and confirm the import.

Each selected worksheet becomes an independent dataset named Workbook - Worksheet. If that name already exists, JASC adds its normal numeric suffix. Every imported dataset retains the workbook as its source path.

Excel import is intentionally value-only:

- Cached worksheet cell values are imported. JASC does not execute Excel formulas; formulas without a saved cached result may appear blank.
- VBA macros, queries, external connections, and external links are never executed or refreshed.
- Charts, pivot layouts/caches, images, shapes, controls, comments, named ranges, filters, totals rows, formatting, conditional formatting, validation, protection, merged-cell layout, print settings, and display settings are not reproduced.
- Cells backing a chart, pivot result, or Excel table can still be imported when they are present in a selected worksheet grid.
- One malformed or missing selected tab is reported without discarding other successfully parsed tabs. Empty tabs and worksheets declined at the large-data warning are skipped.

Excel workbooks are compressed containers, so their in-memory DataFrames can be substantially larger than the file on disk. The normal file-size and per-worksheet memory warnings apply.

Create a blank worksheet

Choose File > New Blank Sheet (Ctrl+N), the toolbar action, or New Sheet in the dataset navigator. JASC creates a normal workspace dataset with:

- 25 initially blank rows.

- Auto-typed x and y columns. All-numeric input is stored numerically, while dates and text remain accepted without changing metadata first.
- X and Y column roles already assigned.

The sheet is immediately editable in the Data tab. Use Data menu commands to add or insert rows and columns.

Fill or create a numeric column

Choose Data > Fill Range... to populate values. The dialog now makes the destination explicit:

- Target action: update a selected existing column, or create a separately named new column.
- Fill pattern: an arithmetic sequence, or one constant value repeated deliberately.
- Start row: one-based worksheet row.
- Count: number of values.
- Start value and Step.

The Result line previews the action, target, affected row interval, and generated values before the operation runs. Launching Fill Range from a selected column defaults to updating that column; without a column selection it defaults to creating a new column. JASC rejects a new-column name that already exists instead of silently changing the operation. For a sequence it writes start + offset * step; Constant value is the same operation with an explicit zero step. JASC grows the worksheet if required and converts a blank/object-backed destination to a real numeric column, so it can immediately be used by formulas such as sin(col1). Existing text outside the replaced rows is rejected before any write; replace it or create a separate numeric column. JASC marks the output Numeric, assigns the X role when the name is x, time, or t, removes a stale formula from an overwritten column, and records provenance in the column comment.

Smart Fill from existing cell values

Use Data > Smart Fill Selection..., the Smart Fill inspector button, or Columns > Smart Fill Selection... in the worksheet context menu to complete blank cells from values already entered in the sheet:

1. Enter seed values when the selected mode requires them.
2. Select one contiguous rectangle containing the seeds and blank destination cells. A Custom List may instead fill a completely blank selection.
3. Open Smart Fill and select Auto, Numeric, Date-Time, Category Cycle, or Custom List.

Smart Fill reports the existing-anchor count for each selected column, treats every existing value as an anchor, and writes only blank selected cells. It handles:

- Auto chooses Numeric when every anchor is a finite number, Date-Time for datetime values or recognizable date/time strings, and Category Cycle for other text. A column explicitly marked Text remains categorical in Auto even if its labels look numeric.
-

Numeric repeats one seed or applies linear interpolation/extrapolation to two or more anchors using visible worksheet row positions. For example, 1, 2, blank, blank becomes 1, 2, 3, 4; 1, blank, 2 becomes 1, 1.5, 2. Numeric pattern controls can also detect or explicitly select two through eight interleaved linear sequences: 1, 10, 2, 20, blank, blank becomes 1, 10, 2, 20, 3, 30.

- Date-Time repeats one timestamp or infers the interval between two or more anchors. Numeric text dates accept every ordering of day, month, and year (day/month/year, month/day/year, year/month/day, year/day/month, day/year/month, or month/year/day). Smart Fill infers one order from all anchors, uses sequence evidence when day and month are both 12 or less, falls back deterministically to day-first if the anchors cannot distinguish them, and keeps the inferred order, separator, and zero-padding in generated cells. Exact same-day or month-end monthly/yearly patterns use calendar offsets, so 2024-01-31, 2024-02-29 continues with 2024-03-31; other patterns use elapsed-time interpolation/extrapolation.
- Category Cycle takes the first consecutive seed block as one complete cycle. Red, Blue, Green, blank, blank continues with Red, Blue. Additional nonblank anchors must agree with the cycle.
- Custom List accepts one ordered item per line. Existing anchors align the list case-insensitively; with no anchors, filling starts at its first item. Lists can represent weekdays, months, cities, colours, sample states, or a user-specific order.
- Several selected columns are inferred independently using the same requested mode and custom list.

The operation is deliberately deterministic and does not query the web. It does not guess arbitrary semantic relationships, nonlinear numeric models, or worksheet cell background formatting: colour names are category values, while actual cell colours would require a separate formatting system. Smart Fill is undoable, invalidates stale whole-column formula metadata in modified columns, and records its resolved mode and pattern in column provenance.

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5. Digitizing Data from Figure Images

Use File > Import From Figure... to extract calibrated data from PNG, JPEG, BMP, TIFF, or GIF figures. First choose the image type:

- XY graph calibrates graph axes and extracts curves, lines, or symbols.
- 2D colour plot / heatmap calibrates X/Y plus a displayed continuous colour scale and returns clicked X/Y/Z probes.

Both workflows share image loading, optional perspective correction, linear/log X/Y calibration, canvas navigation, and normal workspace-dataset output.

Canvas controls and tool state

The toolbar above/below the image provides Load Image, Fit View, 1:1, Zoom In, Zoom Out, Pan, and Magnifier. The mouse wheel zooms around the pointer. The Precision Zoom panel follows the cursor and shows an enlarged neighborhood with pixel and calibrated-coordinate readouts.

The yellow mode banner reports what the next canvas click or drag will do. It is a status display, not a button. Use the controls in the Correction, Calibration, and Extraction tabs to activate a picking tool. Enable Pan when dragging should move the image without picking data.

Step 1: optional perspective correction

Use the Correction tab when a photograph or scan is skewed.

1. Click Pick top-left, Pick top-right, Pick bottom-right, and Pick bottom-left, clicking the corresponding plot-frame or axis-bar corners in that order.
2. Review the green quadrilateral.
3. Set the corrected image width and height in pixels.
4. Click Apply Correction.

Calibration and extraction then operate on the rectified working image. Clear Points removes the four correction points. Reset Original restores the original image and clears pixel-dependent calibration and extraction state.

Correction is optional. When you switch from Correction to Calibration, JASC automatically changes the canvas tool from the correction corners to the first unpicked X/Y calibration reference. Returning to Correction similarly selects the first unpicked correction corner.

Step 2: calibrate X and Y

The calibration maps pixels to data coordinates using two references per axis.

1. In Calibration, select X axis point 1, click its pixel location, and enter the known X value.
2. Repeat for X point 2, Y point 1, and Y point 2.
3. Choose Linear or Log10 independently for X and Y.
4. Review the calibration status. Use Clear Calibration to start over.

Calibration values must be distinct on each axis. Log10 calibration requires positive reference values and is appropriate only for logarithmic axes.

XY graph: Step 3A, manual point extraction

1. Open Extraction and click Manual Mode.
2. Click data points on the image. Use Stop Picking when finished.
3. Use Undo Last to remove the latest manual point or Clear Manual to remove all manual points.

To attach uncertainty to a manual point:

1. Select the target point number in Error target point.
2. Click Radius Error, then click one radial boundary. This assigns symmetric pixel uncertainty from the distance to the point.
3. Alternatively, click 4-Point Error and pick the X-low, X-high, Y-low, and Y-high boundaries.

Manual uncertainties appear as blue error bars and are exported in both pixel and calibrated data units.

Four-point picks retain their separate graphical endpoints in x_lower_bound, x_upper_bound, y_lower_bound, and y_upper_bound, including asymmetric and logarithmically calibrated bounds. The original symmetric maximum-width columns remain for compatibility. Digitized geometry has no inferred percentile coverage; specify the probability levels and centre meaning before using it for fitting.

XY graph: Step 3B, automatic curve or line tracing

The Automatic Extraction Region is shared by curve tracing and symbol detection.

1. Optionally click ROI Brush and paint the region to scan. Adjust Brush size as needed. Without ROI marks, JASC scans the full working image. X/Y calibration points define the coordinate scale only; they do not crop automatic extraction. Pixel-to-data conversion extrapolates outside the calibration references when necessary.
2. In the Curves/Lines subtab, choose Dark pixels or Selected color.
3. For dark pixels, adjust Dark threshold.
4. For color tracing, use Pick Color... or Sample From Image, then set Color tolerance.
5. Set X pixel step to control horizontal sampling density.
6. Click Run Auto Trace. Use Clear Auto to discard the trace.

XY graph: Step 3C, symbol detection

The Symbols subtab has two related paths:

- Detect Symbols finds connected foreground components using the current dark/color rule and ROI. Use Min area, Max area, and Max symbols to reject lines, noise, or oversized objects.
- Detect Sampled matches a learned marker template. Click Sample Symbol, then click near a marker center for radius-based capture or drag a box around it. Configure the shape prior, sample radius, minimum match score, minimum separation, and maximum symbol count before detection.

Shape priors are Auto from sample, Dot / circle, X cross, Plus cross, Square, Diamond, Triangle up, and Triangle down. They help suppress line pixels touching a marker. Clear Template discards the learned template; Clear Symbols discards detected centers.

Symbol outputs include area, width, height, match score when applicable, template dimensions, and uncertainty estimated from marker size.

XY graph: Step 4, review and create one or more datasets

Open Output to set the dataset name and inspect the read-only preview. Each row can include:

- Calibrated X, Y, dX, and dY.
- Pixel X, Pixel Y, dPx, and dPy.
- Extraction method, trace rule/color, ROI count, and source image.
- Symbol ID, area, score, size, and template diagnostics.

To create a worksheet and continue extracting from the same figure without repeating calibration:

1. Enter a distinct dataset name for the completed curve.
2. Click Create Dataset & Continue. The worksheet is added to the workspace immediately while the digitizer remains open.
3. After the worksheet is successfully added, JASC clears only curve-specific manual points, uncertainty marks, automatic trace, symbol results/template, and ROI marks. The embedded working image, perspective-correction result, and X/Y calibration remain available. If worksheet creation is declined or fails, the current extraction remains intact.
4. Extract the next curve and repeat Create Dataset & Continue for every additional worksheet.

Queue Current Curve & Start New remains available when several completed curves should be imported together when the dialog closes. The Output status distinguishes queued curves from datasets already created. Close warns before discarding an uncreated current extraction. Loading another image, applying a new perspective correction, or resetting to the original image changes pixel geometry, so JASC asks for confirmation before clearing queued curves and calibration.

JASC requires a loaded image, complete four-point calibration, at least one manual, traced, or symbol point, and a dataset name. Each created dataset receives numeric coordinate metadata, X/Y/error roles where appropriate, source/provenance comments, the image source path, and a resumable figure-session snapshot.

Resume a saved figure extraction

Every dataset created through Create Dataset & Continue stores a reference to its figure session. Choose the dataset and use Data > Resume Figure Extraction... or the dataset context menu to reopen it. JASC restores:

- The exact working image used for digitization, embedded in the .jasc project rather than loaded from the original external path.
- X/Y calibration pixels, values, and linear/log scale choices.
- Whether the embedded working image resulted from perspective correction.
- For 2D colour plots, colour-bar endpoints and values, normalization, sampling controls, validity threshold, and probe patch radius.

Extracted point markers are deliberately not restored: reopening starts a clean next curve/probe set on the retained image and calibration. Datasets created from the same session share one embedded image inside the project archive. Saving embedded raster images increases .jasc project size, particularly when unrelated figures or high-resolution images are used.

2D colour plot / heatmap workflow

Choose 2D colour plot / heatmap in the initial figure-type dialog. This workflow learns the palette empirically from the colour bar as it actually appears in the image, rather than assuming a named Matplotlib colour map.

1. Load the image and optionally apply perspective correction.
2. Calibrate X and Y exactly as described above. Reference points may be inside the full data region; the mapping extrapolates beyond them.
3. In Displayed Colour Scale, enter the numeric values represented at the two ends and choose Linear or Log10 value normalization. Log10 endpoints must be positive.
4. Click Pick Low End, then click the centre of the colour bar at the entered low-value end. Click Pick High End and click its high-value end. Endpoints may be horizontal, vertical, diagonal, or reversed in screen direction.
5. Adjust Bar sample width if necessary. JASC samples a median stripe across that width along the complete endpoint-to-endpoint segment. Palette samples controls its resolution. Click Rebuild Scale and inspect the displayed sampled palette.
6. Set the probe Median patch radius. A small median patch reduces isolated compression noise, but a large radius can mix neighbouring colours near sharp boundaries.
7. Click Start Probing, then click locations in the colour plot. The shared magnifier shows live X, Y, estimated Z, RGB, confidence, and validity. Click Stop when finished.
8. In Output, inspect the rows. Use Undo Last, Remove Selected, or Clear Probes to clean the result. Enter its dataset name and click Create Dataset & Continue to add the worksheet immediately while retaining the image, X/Y calibration, and colour-scale calibration. Only the probes are cleared, so additional probe datasets can be created without closing or recalibrating.

Each output row contains calibrated x, y, and estimated z; source pixel coordinates; median red/green/blue values; normalized palette position; CIE Lab colour distance; confidence; ambiguity and validity flags; patch radius; extraction method; and source image. X, Y, and Z metadata roles are assigned automatically. Rows that fail the distance threshold or are ambiguous are deliberately retained so they can be inspected and filtered instead of silently discarded. Maximum Lab distance controls the validity cutoff.

> Accuracy disclaimer: colour values are estimates from rendered raster pixels. Lossy image compression, anti-aliasing, resizing, overlaid text or contours, transparency, and limited palette resolution introduce inherent error. Always review colour distance, confidence, ambiguity, and validity before scientific use. Prefer an original lossless image and probe away from labels, grid lines, and boundaries when possible.

The current colour workflow supports continuous empirical colour bars and clicked point probes. Discrete/categorical palettes and automatic full-grid reconstruction are not yet implemented.

---

6. Editing Worksheets and Column Metadata

Select and edit cells

- Double-click a cell, enter a value, and press Enter to commit.
- Blank input is stored as a missing value.
- Numeric/Scientific metadata rejects non-numeric text. Text columns preserve entered text.
- Numerical operations also inspect Auto columns: blanks may be treated as missing where

supported, but nonblank text is never silently discarded. The error identifies the
first visible row and an example value so the worksheet can be corrected before retrying.

- Select complete rows or columns using their headers; use Ctrl/Shift selection where supported by the platform.

Add, insert, and delete structure

- Data > Add Row appends a blank row.
- Data > Insert Row inserts before the active/selected row.
- Data > Add Column... appends a uniquely named blank column.
- Data > Insert Column... inserts before the active/selected column.
- Data > Resize Sheet..., the Resize Sheet inspector button, or the worksheet context menu sets the total row and column counts together. Growing adds empty cells and uniquely named columns. Shrinking asks for confirmation because values, formulas, and metadata outside the new limits are removed.
- Edit > Delete Selected Rows or Shift+Delete removes every selected row after confirmation.
- Edit > Delete Selected Columns or Ctrl+Delete removes every selected column after confirmation.
- Delete clears values in the selected cells without removing worksheet structure. The clear is undoable and invalidates stored formula metadata for affected columns because their generated result is no longer intact.

Row and column edits participate in Undo/Redo when the worksheet fits within the history budget.

Clipboard operations

1. Select cells and press Ctrl+C, or right-click and choose Copy Values.
2. Activate the destination cell and press Ctrl+V, or choose Paste Values.

Values use tab-delimited clipboard text. JASC expands the worksheet to fit the pasted rectangle, infers numeric blocks where safe, preserves explicit Text destinations, and clears formulas on overwritten columns.

Right-click and choose Copy Row(s) to copy every column from all rows touched by the selection. To exchange rows and columns, choose Paste Values Transposed or press Ctrl+Shift+V. The complete rectangular clipboard matrix is transposed: an M-row by N-column copy becomes N rows by M columns, so this works for single rows, single columns, and multi-row/multi-column selections.

Right-click selected columns and choose Copy Column(s). In another JASC worksheet, right-click a destination column and choose Paste Column(s). Whole-column copy uses a JASC-specific clipboard payload that preserves:

- Column headers.
- Values.
- Column roles, types, precision, units, long names, and comments.

Pasted columns are inserted before the destination and renamed automatically if their names collide. The metadata-preserving payload is available only while the clipboard still contains the JASC column data; plain-text applications receive the tabular values.

JASC validates every metadata-preserving pasted column before inserting any of them. If a Numeric/Scientific column contains nonnumeric text, the whole paste is rejected and the worksheet remains unchanged.

Undo and redo

Use Edit > Undo (Ctrl+Z) and Edit > Redo (Ctrl+Y). History covers cell edits and clearing, worksheet resizing, row/column changes, clipboard pastes, Fill Range, Smart Fill, formula writes, metadata/column renames, and current-sheet calculus/smoothing changes where the operation captures editor state.

History is dataset-specific and resets when the displayed dataset changes. Each undo and redo stack retains at most 20 snapshots and approximately 256 MiB. If one worksheet snapshot exceeds the budget, JASC performs the edit but warns that it cannot be undone. Creating/removing datasets or batch changes to non-visible sheets are not guaranteed to be undoable from the current table history.

Column Metadata dialog

Open Data > Column Metadata..., double-click a column header, or use the table context menu. Each row exposes:

Field	Purpose
Column name	Renames the column; names must be nonblank and unique
Role	Hints default selections in plotting, fitting, interpolation, and analysis
Type	Controls display and explicit numeric/text validation
Precision	Auto or 0-15 displayed decimal places; Scientific uses exponential notation

Unit	Included in suggested plot-axis labels
Long name	Human-readable axis/header description
Comment	Notes, source details, and generated-operation provenance

Available roles are None, X, Y, Z, Error, X error, Y error, Group, Label, and Weight. Available types are Auto, Numeric, Scientific, and Text.

Changing display precision does not change stored values. Renaming a column carries its formula and metadata forward and updates active and saved plot column references.

Formula and provenance indicators

Formula-backed and metadata-rich column headers expose details in tooltips. Generated operations append an Operation: line to column comments, replacing the prior generated-operation line while retaining other user comments.

---

7. Data Preparation

Data Preparation commands create new result datasets. The source sheet remains unchanged. Matching metadata is copied where output columns retain their original names, and JASC records the operation in output-column comments.

If an operation returns no rows, JASC asks whether to create the empty output sheet.

Extract Subset

Use Data > Extract Subset..., the Data Prep inspector, or a worksheet's Data Preparation context submenu to create a positional sample of the active sheet. Row and column rules are independent and can be combined:

- All positions keeps that dimension unchanged.
- Odd positions keeps visible worksheet positions 1, 3, 5, and so on.
- Even positions keeps visible worksheet positions 2, 4, 6, and so on.
- Every Nth position keeps the selected starting position and advances by N. For example, interval 3 starting at row 2 keeps rows 2, 5, 8, and so on.

The dialog previews the resulting row and column counts before creation. Positions are based on the displayed 1-based worksheet numbering, not the internal Pandas index. The source remains unchanged, output rows are renumbered from zero internally, retained columns keep their metadata, and the row/column rules are written to the result's operation provenance.

Use Filter / Query instead when rows should be chosen by their values rather than their positions.

Filter / Query

Choose one dataset, one column, one condition, and an output sheet. Supported conditions are:

- Equals / does not equal.
- Greater than, greater than or equal, less than, less than or equal.
- Between two numeric values, inclusive.
- Contains / starts with.
- Is in a comma-separated list.
- Is blank / is not blank.

Numeric comparison text must parse as a number. The current dialog applies one condition at a time; multi-condition query building and in-place filtering are not implemented.

Mean/median/interpolation cleanup, numeric filters, grouping, and binning reject nonblank nonnumeric source cells instead of treating them as missing. Fill with value also refuses a text replacement for a numeric or safely inferred numeric column; mark an

intentional category column Text when textual missing-value labels are required.

Sort

Select one or more columns, Ascending or Descending, and an output name. Multiple keys are applied in their worksheet order using a stable sort.

Remove Duplicates

Select the columns that define a duplicate group and choose:

- Keep first occurrence.
- Keep last occurrence.
- Remove every row in a duplicate group.

Blank/NaN values in the same selected key columns are treated as matching.

Missing Values

Select one or more columns and a strategy:

- Drop rows missing a selected value.
- Fill with a literal value.
- Fill numeric blanks with the column mean.
- Fill numeric blanks with the column median.
- Interpolate numeric gaps in both directions.

Mean, median, and interpolation coerce the selected output columns to numeric values; non-numeric entries become missing before filling.

Bin / Average

Choose a numeric bin column, one or more value columns, and a bin count. JASC creates equal-width bins between the observed minimum and maximum and reports:

- Bin interval, start, end, and center.
- Row count.
- Mean, sample standard deviation, SEM, minimum, and maximum for every selected value column.

Binning requires a non-constant numeric bin column.

Group / Aggregate

Choose one or more group columns, distinct numeric value columns, and one or more aggregations: count, mean, standard deviation, SEM, minimum, maximum, and sum. Missing group keys are retained as groups.

Merge / Join

Choose left and right datasets, one key from each, and Inner, Left, Right, or Outer join. Same-named right-side columns receive a _right suffix where required.

Reshape

- Transpose converts columns to rows and adds a source_column field.
- Melt / stack retains selected identifier columns and stacks selected value columns into variable and value fields.

---

8. Formulas and Expressions

Choose Data > Apply Formula... (Ctrl+Shift+F) to calculate a new column or update an existing column with a constrained Python-like expression.

Basic workflow

1. Select or type a Target column. When the dialog was launched from one selected existing column, that column is preselected as an update target.
2. Choose the explicit Output action:
- Create a unique new column.
- Stop and warn.
- Overwrite/update the existing column.
3. Enter the expression.
4. Choose the Operation scope:
- Entire target column evaluates and writes every row. This is the mode used for multi-sheet batches.
- Selected cells in target column writes only cells that are selected in that specific target column. It applies only to the current sheet and always updates the existing target column.
5. Select one or more sheets for an entire-column/batch operation.
6. Choose exact-name or position column matching for batch targets.
7. For formula batches, choose Auto, Sequential, or Parallel processes, plus the maximum worker count.
8. Confirm the preflight preview and run.

When a formula-backed target column is selected, JASC reloads its stored expression unless doing so would overwrite an expression currently being edited.

Referencing columns and rows

Syntax	Meaning
voltage	A column whose name is a safe Python identifier and does not conflict with a built-in name
col1, col2, ...	First, second, and later columns by position
col("Voltage (V)")	Exact-name access for spaces, punctuation, keywords, or conflicting names
sheet("Calibration", "Gain")	Column Gain from worksheet Calibration, aligned by visible row position
row	One-based row-number series
index	The pandas index as a series

Use the worksheet/column selectors and insert buttons to avoid typing mistakes. Choosing the current worksheet inserts colN or col("Name"); choosing another worksheet inserts a complete sheet("Worksheet", "Column") reference.

Cross-worksheet values are matched by visible row position, independently of the pandas index. If the referenced worksheet is shorter than the target, remaining target rows receive missing values; extra source rows are ignored. Renaming a worksheet updates literal sheet(...) references saved in formula metadata.

Examples:

```
col("Voltage (V)") / col("Current (A)")
col("Raw signal") * sheet("Calibration", "Gain") + sheet("Calibration", "Offset")
np.sin(col1) * np.exp(-col2)
where(col1 > 0, sqrt(col1), nan)
row * 0.1
special.erf((col1 - 5) / 2)
signal.savgol_filter(col2, 11, 3)
np.random.normal(0 * col1, 1)
```

Available safe namespaces

The evaluator exposes a curated subset, not unrestricted NumPy/SciPy or Python.

- NumPy: functions are available both directly (sin, sqrt, where) and through np (np.sin, np.sqrt, np.where). Categories include arithmetic, trigonometry, logarithms, rounding, reductions, cumulative operations, gradients/differences, clipping, conditions, interpolation, finite-value checks, and array helpers.
- NumPy subnamespaces: selected np.fft functions, np.linalg.norm, and selected np.random generators.
- SciPy aliases: scipy and sp.
- SciPy subnamespaces: special, signal, stats, and integrate, also accessible as scipy.special, scipy.signal, scipy.stats, and scipy.integrate.
- Constants: pi, e, nan, and inf, also available through np.

The Help panel inside Apply Formula lists the allowed functions and examples for the installed version.

Safety and expression rules

- Use ** for powers. JASC rejects ^ because Python interprets it as bitwise XOR.
- File access, imports, Python builtins, private attributes, assignments, comprehensions, and arbitrary object access are blocked.
- A scalar result such as 5 or np.mean(x) is deliberately repeated across the destination rows. Reference a column, such as x * 2, for row-by-row results. A vector result must match the worksheet length.
- Wholly numeric object/string columns are promoted before formula evaluation, so imported

values such as "1", "2" work numerically. Mixed number/text columns cannot silently perform string repetition during numerical expressions; correct the source data or use an explicit text expression.

- Numeric/Scientific formula targets reject textual results before assignment. Explicit Text targets receive string values, and Numeric/Scientific metadata is also validated before every downstream math operation.
- Entire-column application stores the expression in project metadata and records it in the target column comment. Selected-cell application records cell-scoped provenance but removes any whole-column formula marker because the complete column no longer represents one formula.

---

9. Calculus, Smoothing, FFT, and Interpolation

All workbenches in this section support multiple compatible worksheets. Derivative, Integral, and Smoothing write columns into target worksheets; FFT, Inverse FFT, and Interpolation create result datasets.

Target signal/value lists start with no columns selected. Choose one or more columns explicitly in the dialog before applying the operation; each click toggles a column, so selecting two signals does not require accepting an automatically chosen first signal. JASC may still suggest the coordinate/X column from metadata because it defines spacing or alignment, but it never silently selects the Y, signal, spectrum, interpolation-value, statistics, or curve-fit response columns. Optional fit uncertainty/weight columns also start empty. Applying without the required selection shows a validation message and performs no operation.

Derivative

Choose Analysis > Derivative..., select one X column and one or more Y columns, then choose:

- NumPy gradient.
- Forward difference.
- Backward difference.
- Central difference.
- Savitzky-Golay derivative.

Set a target column or template, conflict policy, matching mode, target sheets, and whether to plot immediately. Use {column} in the target name when processing several Y columns.

X/Y data are aligned after numeric conversion and blank removal. X values cannot contain duplicate positions. The Savitzky-Golay derivative also requires evenly spaced X values.

Integral

Choose Analysis > Integral.... Methods are:

- Cumulative trapezoid.
- Cumulative Simpson.
- Left rectangle.
- Right rectangle.
- Midpoint rectangle.

The Initial value is the integration constant added to the cumulative result. Integral has the same multiple-Y, target, conflict, batch, and immediate-plot controls as Derivative.

Smoothing

Choose Analysis > Smoothing.... The dialog previews the first selected Y signal and limits preview rendering to at most 4,000 evenly sampled points.

Method	Relevant controls
Adjacent average	Window points
Moving median	Odd window points
Savitzky-Golay	Odd window points and polynomial order smaller than the window
LOWESS / LOESS	Fraction of data and robust iterations
FFT low-pass	Cutoff as percent of Nyquist; evenly spaced X required
Derivative smoothing + integration	Derivative smoother, optional odd slope-spike median prefilter, endpoint preservation

Derivative reconstruction calculates dy/dx, optionally removes derivative spikes, applies any non-recursive smoothing method, then integrates from the original first Y value. Preserve first and last data values distributes residual drift so the reconstruction ends at the original last value.

The default target template is {column}_smoothed. Choose a conflict policy, name/position matching, target sheets, and whether to plot original versus smoothed data immediately.

FFT

Choose Analysis > FFT... and select one or more signal columns. Choose an X/time column to derive sample spacing, or Sample index for spacing 1. Available windows are:

- None / rectangular.
- Hann.
- Hamming.
- Blackman.
- Bartlett.
- Flat top.

When an X/time column is used, values must be finite and evenly spaced within the operation tolerance. Each result dataset contains frequency, real, imaginary, magnitude, and phase. The optional immediate plot shows magnitude versus frequency.

Inverse FFT

Choose Analysis > Inverse FFT... and select one or more real spectrum columns. Imaginary input may be:

- Empty, meaning zero imaginary component.
- One column reused for every selected real column.
- The same number of columns as real selections, paired by selection order.

Each result contains sample, real, imaginary, and magnitude. The optional immediate plot shows the reconstructed real signal.

Interpolation

Choose Analysis > Interpolation.... Select one or more value columns and a dimensional mode.

Mode	Methods
1D	Linear, nearest neighbor, cubic spline, PCHIP, Akima, smoothing spline
2D	Nearest neighbor, linear scattered, cubic spline, piecewise cubic / Clough-Tocher, smooth bivariate spline, RBF linear, thin plate, cubic, Gaussian
N-D	Nearest neighbor, linear scattered, RBF linear, thin plate, cubic, Gaussian

N-D mode requires at least two unique coordinate columns. Value columns must differ from coordinate columns. Duplicate source coordinates are averaged.

- Generated range/grid: available for 1D and 2D. Set start, end, and count for each dimension. A 2D query produces the Cartesian X/Y grid.
- From existing columns: choose one query column per coordinate dimension.
- Manual coordinates: enter one point per line, separating values with spaces, commas, or tabs.

Generated queries are limited to 1,000,000 points.

- Outside range applies to 1D: blank outside the source range, extrapolate where supported, or clamp to edge values.
- Smoothing is enabled for smoothing splines and RBF methods.
- RBF epsilon is enabled for Gaussian RBF; zero displays as automatic.

The result preserves query coordinates and adds one {value_column}_interpolated prediction column per selected value. Result-sheet templates can use {source} and {column} in batch mode.

Baseline Correction and Peak Analysis

Choose Analysis > Baseline Correction... for a correction-only workflow, Analysis > Peak Analysis... for a general spectrum workflow, or **Analysis > Raman Analysis...** for the same tested configuration and numerical engines with results presented in the dedicated Raman Analysis module. The Raman entry does

not maintain a divergent fitting implementation.

The worksheet from which the command was launched is preselected, but it is not locked: choose any workspace worksheet in Source & Range, then select one X/spectral-axis column and one intensity column explicitly. The engines accept ascending or descending and evenly or irregularly spaced X values.
Missing rows remain missing, while contiguous finite segments are analyzed independently. An optional physical-X range can limit the analysis.

Preview Baseline, Detect / Refresh Peaks, and Fit Selected Components calculate in the background. You can keep editing controls while a preview runs. Changes to source, pixel, range, or algorithm settings invalidate outdated results immediately; typing pauses for 250 ms before the latest update starts. Only the newest pending request is kept. Native baseline/detection phases finish before a replacement starts; composite fits also check cancellation during optimization.

Use Stop Preview to cancel pending/current preview work while keeping the dialog open. Editing fit models or reviewed peaks cancels an outdated fit without replacing the review table. Manual additions, repositioned/disabled peaks, removed candidates, and component model choices survive refreshes for the same source and pixel within the selected range. Changing the source/pixel or loading a parameter profile starts a new review.

OK uses the current review and completes any required preview/fit before accepting. If you change inputs while acceptance is pending, the dialog stays open for review; click OK again when ready. Cancel, Esc, closing the dialog, or replacing/closing its project cancels previews and waits safely for active work to stop. Preview drawing is reduced to at most 6,000 points per curve; numerical results and exports keep their full resolution.

For a normalized long-form map, choose Long-form map (group by X/Y) under Source layout. JASC selects this automatically when the worksheet contains multiple finite X/Y coordinate groups, but the control remains explicit so a worksheet can be interpreted deliberately. Choose one entry under **Preview X/Y pixel**. Only that pixel's spectrum is used for interactive baseline, detection, and component-fit preview; the complete repeated-axis table is never treated as one spectrum.

After the representative pixel looks correct, choose **Current long-form Raman map (group by X/Y) under Apply to**. Baseline Correction then applies the reviewed baseline parameters independently to every X/Y group. Peak Analysis does the same for baseline/detection/fitting and matches reviewed components by center tolerance. The saved analysis retains X and Y on every metric, curve, peak, and parameter row. A single-pixel result remains available by choosing Current preview spectrum / selected pixel instead.

Baseline-only map correction runs in the background after accepting the dialog. The status bar shows completed/total pixels and Cancel. Successful pixels form one ordinary corrected worksheet with raw, baseline, and corrected signal columns; Worksheet Metadata records requested/successful/failed pixel counts and error details. An individual baseline failure skips that pixel and lets the other pixels continue. If every pixel fails, the run reports an error without creating a worksheet. Numeric/text validation remains strict across all selected columns.

Cancelling discards the entire pending map, including pixels already corrected. Changing the source data, column metadata, or project discards pending results. Closing the window requests cancellation and waits for the active worker to stop. A native baseline call finishes before cancellation takes effect. Inputs and assembled results remain in memory.

The Baseline tab provides:

- arPLS (automatic default), AsLS, airPLS, SNIP, and rubber-band methods;
- manual anchors with linear, PCHIP, or Akima interpolation. Choose **Explicit X:Y baseline points to enter x:y pairs separated by semicolons, or X positions (sample each source signal)** to enter a list such as 100; 250; 400; 700;
- constant, linear, and polynomial backgrounds; and
- explicit positive, negative, or mixed peak polarity.

X-only anchors are the recommended reusable segment definition. JASC stores the physical X positions, then independently interpolates the current source signal at those positions whenever the correction runs. The same profile therefore uses identical spectral locations across worksheets or map pixels without incorrectly copying one spectrum's baseline Y values into another. Every anchor must be unique and inside the finite, selected X range. The resolved X:Y points are retained in operation provenance.

pybaselines is the preferred backend. If it is absent, JASC uses a sparse, memory-bounded fallback and reports that fact in the warnings/provenance. The preview reports convergence and iteration diagnostics. One-sided baseline methods warn when mixed-polarity peaks may bias the result.

The Detection tab uses prominence-based SciPy peak finding on the corrected signal. Prominence may be explicit or estimated from robust first-difference noise. Height/SNR, minimum center distance, minimum/maximum FWHM, minimum plateau width, maximum candidate count, polarity, and optional moving-median or Savitzky-Golay smoothing are configurable. Distances and widths are always in physical X units, including on irregular axes.

The preview shows raw data with its baseline and corrected data with linked peak markers. In the review table you can disable a candidate, edit its center, remove it, or add a candidate at a physical X coordinate. Numerical analysis

always uses the complete input; only preview rendering is decimated for long spectra.

Use the toolbar directly above the preview to inspect small features. Zoom lets you drag a rectangle around a small peak, Pan moves through the enlarged view, Back/Forward revisit view changes, and Home restores the complete preview. The toolbar's save action exports the currently displayed preview image; it does not create an analysis worksheet or parameter profile.

Reusing Peak Analysis parameters

Use Save Parameters... and Load Parameters... at the top of either Baseline Correction or Peak Analysis. Profiles are portable, human-readable UTF-8 JSON files with the .jasc-analysis.json extension. A profile saved from Baseline Correction contains the range and baseline subset; one saved from Peak Analysis additionally contains detector and fit settings. Together the fields are:

- the optional physical analysis range;
- every baseline method control, interpolation choice, and explicit or X-only anchor list;
- every detection threshold, physical distance/FWHM/plateau control, width relative height, smoothing choice, and candidate limit;
- corrected/raw fit mode, the shared baseline, loss function, maximum evaluations, and peak-center matching tolerance; and
- reviewed component enable states, reference centers, polarities, and registered peak-model choices.

Loading a profile changes algorithm settings only. It does not change the source worksheet, source columns, output names, output scope, or selected batch sheets. After detection refreshes, saved component models are assigned to the nearest compatible reviewed centers. Batch/map execution continues to enforce the explicit center-matching tolerance. Profile files carry a schema name and version; unknown or malformed profiles are rejected instead of being partially guessed.

The Fit Components tab creates one row per reviewed peak. Enable or disable each component and independently choose a registered Gaussian, Lorentzian, Voigt, pseudo-Voigt, Pearson VII, sech-squared, Laplace, Fano, asymmetric Gaussian, split Lorentzian, exponentially modified Gaussian, Fraser-Suzuki, or Doniach-Sunjic model. Starting centers, heights/areas, and physical widths come from the reviewed detector results with model-specific conversions.

Fano is handled as an interference resonance rather than an ordinary one-lobe peak. Its positive amplitude is the continuum-referenced scale, center is the unshifted resonance position, gamma is half the natural linewidth, and signed

q sets the peak/dip orientation. JASC tries several physically distinct q starts and reports resonance position, natural linewidth, q, peak position, dip position, peak contrast, dip contrast, and propagated position/intensity errors. Its integrated area is explicitly a signed integral over the selected fitting window and should not be interpreted as a window-independent total area.

Use Corrected signal to fit the baseline-corrected values with a fixed zero fit baseline. Use Raw signal to fit one shared zero, constant, linear, or quadratic baseline plus all enabled peaks. Individual peak offsets are suppressed so the baseline is not counted once per component. Least-squares, soft-L1, and Huber loss are available. Fit Selected Components previews the total fit and each component and reports convergence, R squared, RMSE, AIC, and BIC.

Completed full fits are saved first as named analysis sessions under **Workspace > Analyses**. Signal, peak, fit-curve, parameter, and map worksheets are unchecked by default and are optional exports, preventing intermediate results from crowding the Data navigator. Peak rows record source/channel, source row, center, polarity, raw/corrected height, prominence, FWHM, boundaries and bases, signed area, local SNR, plateau width, asymmetry, detector, manual status, and whether the reviewed row is enabled. Worksheet metadata stores the complete baseline/detection settings, convergence, warnings, noise estimate, software versions, and source columns.

The Output & Batch tab can optionally create a composite-fit curve worksheet (observed, shared baseline, total_fit, residual, and every peak component) and a tidy parameter worksheet containing semantic roles, bounds, fixed state, standard errors, integrated areas, fit metrics, status, and errors.

Choose an application scope:

- Current source worksheet fits only the selected source.
- Selected worksheets (batch) applies the reviewed peak models to the checked worksheets. Each spectrum is redetected and matched to the reviewed centers by the selected physical-X tolerance; peaks are never matched only by prominence rank.
- Current long-form Raman map groups the selected worksheet by its normalized X and Y columns and fits every pixel spectrum. Curve, peak, parameter, and summary rows carry their X/Y coordinates. A failed worksheet or pixel is reported in the cumulative summary without discarding successful jobs.

Raman Analysis result viewer

The main workspace starts with only Data and Plot. After a fit is successfully completed, or after a saved result is opened from **Workspace >

Analyses, JASC creates the contextual Raman Results** tab and opens the persistent session there. Merely starting JASC or loading a project does not add an empty Raman tab. The Workspace Analyses section is likewise hidden when the project contains no saved analysis sessions.

The map displays the selected numeric parameter or quality metric over X/Y. Regular grids render as a heatmap and irregular positions as coloured spatial points. Choose All / quality or one reviewed peak, then choose intensity, area, position, FWHM, asymmetry, native parameters/errors, baseline, SNR, RMSE, R squared, or another stored metric. Click a map position, or use **Spectrum / pixel**, to update the linked raw/baseline/corrected spectrum, individual components, and total fit. The primary view deliberately keeps only the 2D map and selected-pixel fit visible so both remain large enough for inspection. The fitted-spectrum title shows only the selected X/Y coordinates rather than repeating a potentially long source filename. Double-click either plot title, an X/Y axis label or tick area, or the fitted-spectrum legend to edit it in place. The axis editor controls its label, scale, automatic/manual limits, label/tick sizes, and grid.

Double-click the spectrum legend, or use Curves / Legend..., to show or hide the raw/original signal, removed preprocessing baseline, corrected signal, fit input, shared fit baseline, total fit, and individual peak components. The same dialog edits legend labels, position, text size, columns, frame, and opacity; the button remains available even if every curve or the legend itself is hidden. Enable Show diagnostics when needed to reveal a compact lower panel with separate Residual / Fit Error, Metric Distribution, and QC Details tabs. Hiding diagnostics returns that space to the map and fitted spectrum.

Use Export Results to Worksheets... only when a normal worksheet is needed. The export dialog independently offers parameter maps/fit quality, all fitted curves, detailed parameters, and detected/reviewed peaks. Published tables retain the analysis and source names in metadata. The internal metrics table contains exactly one row per X/Y pixel. Peak identity follows the reviewed template order (peak_1, peak_2, and so on), not the prominence rank at each pixel. Available map columns include:

- fitted peak_N_intensity, signed integrated peak_N_area, fitted peak_N_position, numerical peak_N_fwhm, and half-width peak_N_asymmetry;
- local peak_N_snr, detected position, model name, every native model parameter, and the covariance-derived standard error for each fitted parameter;
- standardized position/intensity/area/FWHM/asymmetry standard-error columns where that uncertainty is directly available;
- fit_r_squared, fit_rmse, fit_mae, fit_aic, fit_bic, convergence, evaluation count, robust noise estimate, and baseline minimum/mean/maximum; and

- numeric fit_success plus text status/error fields. Failed pixels remain in their spatial row with NaN result cells, preserving the coordinate grid.

The dedicated viewer draws these results directly. For further customization, export Parameter maps and fit quality, then open **2D Maps > Interpolated Heatmap** (or a contour subtype), choose X and Y, and select, for example, peak_1_area, peak_1_position, peak_1_asymmetry, or fit_rmse as Z. RMSE/MAE describe the complete overlapping fit; per-parameter standard errors quantify individual fitted-parameter uncertainty. A derived area or numerical FWHM may have no direct standard error when it is not a native model parameter, in which case its standard-error cell is intentionally blank/NaN rather than estimated without a defensible propagation model.

Peak/Raman batch and map fits run sequentially in a background worker. The status bar shows completed/total spectra and Cancel while the workspace remains accessible. Cancelling discards the entire pending run, including already fitted pixels; no analysis session or optional worksheets are published. Cancellation is checked between preprocessing phases and during optimizer evaluations. A native baseline/detection phase finishes before cancellation takes effect.

Individual failed spectra are recorded in the analysis quality/status table while the remaining spectra continue. When the run completes, its analysis session and requested optional worksheets are published together. Changing the source data, column metadata, or project during fitting discards the pending results. Closing the window requests cancellation and waits safely for the worker to stop.

Input snapshots and result curves remain in memory. Process parallelism and advanced per-parameter bound/fixed/tied editing remain follow-up work.

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10. Statistics Workbench

Choose Analysis > Statistics.... Select an operation, its primary/secondary/group inputs, output name, target sheets, column-binding mode, and optional companion plot. Numeric operations ignore blanks, non-numeric values, and infinities after validating explicit metadata.

Single-column operations

- Complete report.
- Descriptive summary: count, missing count, mean, median, standard deviation, variance, SEM, min, max, range, and coefficient of variation.
- Quantiles and percentiles.
- Distribution shape: skewness and kurtosis.
- Normality tests: Shapiro-Wilk, D'Agostino-Pearson, Jarque-Bera, and Anderson-Darling.
- Outlier diagnostics using Z-score and IQR fences.
- Mean confidence interval.
- Z-score result series.
- Cumulative statistics result series.

Two-column operations

Two-column operations can reuse one secondary column for several primary columns or pair equal-sized selections by order. Paired analyses align rows before dropping incomplete pairs.

- Complete comparison report and pair summary.
- Pearson, Spearman, and Kendall correlations.
- Covariance matrix values.
- Linear regression.
- Error metrics: MAE, RMSE, mean error, and R-squared against the primary column.
- Bland-Altman agreement.
- Paired t-test and Wilcoxon signed-rank.
- Welch independent t-test, Mann-Whitney U, and two-sample Kolmogorov-Smirnov.
- Levene and Bartlett variance tests.

Multiple-column operations

Select two or more columns from the primary sheet:

- Complete multi-column report.
- Descriptive summary.
- Pairwise correlation matrix tests.

- Covariance matrix values.
- Principal component analysis with eigenvalues, explained variance, loadings, and score rows.

Grouped operations

Choose one or more primary numeric columns and exactly one group/factor column, which may come from another sheet:

- Complete grouped report.
- Descriptive summary by group.
- Assumption checks.
- One-way ANOVA.
- Kruskal-Wallis.
- Tukey HSD post-hoc comparisons.
- Pairwise Mann-Whitney comparisons with Holm-Bonferroni adjustment.

Companion plots

Available choices depend on operation scope:

Scope	Plot choices
Single	Auto, histogram + normal curve, box, violin; Z-score/cumulative operations can also plot the result series
Pair	Auto, scatter, scatter + regression line, comparison box, comparison violin, Bland-Altman
Multi	Auto, selected-column box/violin; correlation heatmap for correlation/complete reports; PCA explained variance for PCA/complete reports
Group	Auto, box by group, violin by group

Batch statistics creates long summary results and, when applicable, a wide summary dataset. At most one companion plot is queued for a batch run so a large batch does not replace the plot repeatedly.

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11. Curve Fitting and Custom Models

Standard curve fitting

Choose Analysis > Curve Fit... (Ctrl+Shift+C).

1. Choose the input Worksheet in the Source group, then one X column and one or more Y columns.
2. Select a model category, model, and optimizer.
3. Edit each parameter's initial value, optional lower/upper bounds, and Fixed checkbox.
4. Configure uncertainty, confidence, residual, and output options.
5. Select target sheets and column matching.
6. Confirm the batch preview and run.

The Worksheet selector lists loaded editable sheets and initially uses the current worksheet. Changing it refreshes X/Y, weight and percentile-column choices; response and uncertainty mappings must be selected again. Ordinary single-sheet fitting follows that worksheet automatically. Explicit multi-sheet selections remain under Batch > Apply to sheets, using the chosen source sheet as their column-matching reference. Default output names follow worksheet changes; names you typed are kept.

Built-in models

The Category control keeps the 66 built-in functions navigable. All models is also available when you want the original flat view.

Category	Included models
Polynomial	Constant, linear, quadratic, cubic, quartic, and quintic
Exponential & decay	Exponential growth, one/two/three-component decay, exponential association, and stretched exponential
Growth & sigmoidal	Four- and five-parameter logistic, Gompertz, Richards, Boltzmann, Hill sigmoid, and Weibull saturation
Peak - symmetric	Gaussian, Lorentzian, exact Voigt, pseudo-Voigt, Pearson VII, sech-squared, and Laplace
Peak - asymmetric	Split Gaussian, split Lorentzian, exponentially modified Gaussian, log-normal, Fraser-Suzuki, Doniach-Sunjic, and Fano resonance
Transition & waveform	Error-function and arctangent steps, smooth rectangular pulse, sine, cosine, damped sine/cosine, sine-squared, and two-harmonic Fourier series
Power, logarithmic & rational	Power law, shifted logarithm, shifted reciprocal, and first-order rational

Binding & kinetics	Michaelis-Menten, Hill saturation, substrate inhibition, and one-/two-site binding
Rheology	Bingham, Herschel-Bulkley, Cross, and Carreau
Probability distributions	Normal, gamma, Weibull, and log-normal PDFs plus Gaussian and Weibull CDFs
Special & physical	Bessel J/Y, Heidler impulse, Arrhenius, and wavelength-domain Planck spectrum
Custom	Project-defined safe expression models

The dialog shows the selected category description, equation, parameter order, and static defaults. Width conventions are named explicitly: Gaussian sigma is a standard deviation and Lorentzian gamma is a half width at half maximum. The exact Voigt uses both conventions. The EMG uses area scaling; its center is the mean of the underlying Gaussian component. Planck expects wavelength in metres and Arrhenius expects absolute temperature with activation_over_R in kelvin.

Optimizers and parameter constraints

- Levenberg-Marquardt is suitable for unconstrained least squares.
- Trust Region Reflective is the default and supports bounds.
- Dogbox supports bounded least squares.
- Orthogonal Distance Regression supports uncertainty in both X and Y.

Blank bounds are unbounded. Built-in models prefill recommended domain bounds for scale/width/time parameters and bounded fractions such as pseudo-Voigt eta; these remain editable. Clear all finite bounds before choosing Levenberg-Marquardt, which only supports unconstrained fitting. Fixed parameters remain at their initial values. At least one parameter must remain free.

Uncertainty and weighting

- Select an optional X weight/sigma column.
- Select zero, one, or one-per-Y Y weight/sigma columns.
- Choose Sigma / standard deviation or Weight (larger = stronger).
- Treat sigma as absolute controls covariance scaling.

Selecting an X uncertainty column automatically switches to ODR and locks the optimizer control. Error/sigma values must be positive and finite where used.

Asymmetric percentile bounds

In the Uncertainty tab, enable Use X percentile bounds and/or **Use Y percentile bounds**. Select separate lower/upper endpoint columns and their percentile levels (default 16% and 84%). These are values in the same units as the

central column, not negative/positive error widths. Choose the meaning of the central values and whether the bounds are measurement/sampling or posterior quantiles. Scroll this tab if both axis editors exceed the available height.

Example: a median of 10 with lower endpoint 8 at 16% and upper endpoint 14 at 84% defines a 68% interval with unequal plotted errors of 2 and 4. The probability levels remain explicit; those distances are not automatically standard deviations.

For median measurement values, the method switches to **Percentile likelihood (median model)**. The quantile-normal model exactly matches both selected percentiles and the median. Its CDF smoothly transforms residuals to a normal distribution with Gaussian tails beyond the selected interval; fitting includes the density transformation, not just a rescaled least-squares residual. This is an explicit distribution assumption, since percentile bounds do not uniquely determine a full probability distribution.

With X bounds, independent latent true X positions are fitted jointly with model parameters. A symmetric X sigma can also accompany asymmetric Y bounds, or vice versa. If no Y uncertainty is selected, the likelihood explicitly assumes unit Y standard deviation and reports this. Symmetric and percentile uncertainty cannot both be selected for the same axis. A Y-bound pair currently maps to one selected response column per dialog; the same mapping can run across batch worksheets.

Fitting requires finite, strictly ordered bounds around every retained median. Posterior quantiles and non-median centres are supported for plotting only; they are not silently interpreted as sampling likelihoods. Correlated/shared systematic errors are not represented by this independent-point model.

Generated fit plots retain the asymmetric input bars. Fit/residual worksheet metadata records columns, levels, model version, and negative log likelihood; residuals include original bounds and latent X when applicable. Parameter errors and curve bands use a local Hessian/normal approximation, with fixed parameters retained. A singular/nonpositive curvature or parameter at a constraint makes these local errors unavailable rather than reporting misleading bounds. Bootstrap/profile percentile intervals are not implemented in this slice.

The composite-peak numerical API accepts the same quantile likelihood through uncertainty=QuantileNormal(...). Its dedicated Peak/Raman dialog retains its existing controls and does not yet expose percentile-column mapping.

Outputs and report

Configure:

- Confidence band on/off and confidence percentage, 50.0-99.9%.

- Number of generated fit-curve points, 10-10,000.
- Result sheet name.
- Residual sheet on/off and name.
- Residual plot on the right Y axis.
- Optional long-format parameter covariance/correlation sheet.

The Fit Report includes model/method, parameter estimates, fixed/free state, bounds, standard errors, covariance, R-squared, adjusted R-squared, RMSE, MAE, residual mean/standard deviation, point counts, and warnings. Successful immediate plotting queues raw data, fitted curve, optional confidence band, and optional residuals.

Every successful fit now retains its report in the workspace. A single fit creates <fit-sheet>_report; a batch creates <reference-sheet>_fit_reports with one row per successful fit. Existing sheets receive unique names rather than being overwritten. The table includes full report_text, separate numeric parameter estimates/errors/bounds/fixed states, fit metrics and percentile-setting provenance. Save Project to retain it in .jasc; use normal dataset export for CSV or tab-delimited tables. Failed fits do not create success reports.

The single-fit viewer identifies its saved worksheet and provides Copy Report for plain text, Export Table... for recorded CSV/tab-delimited export, and Open Report Sheet to switch to the saved table. Report creation/export also participates in Python API recording and is recomputed on replay.

Custom fit models

Choose Analysis > Custom Fit Models... to Add, Edit, Duplicate, or Remove project-scoped models.

A custom model defines:

- A unique name.
- An expression using x as the independent variable.
- Comma-separated parameter names.
- One default initial value per parameter.

Example:

```
amplitude * np.exp(-x / tau) + offset
```

Parameter names must be unique Python identifiers, cannot be x, and cannot conflict with built-in model names. The validator exposes a curated selection of NumPy plus scipy.special, scipy.signal, scipy.stats, and scipy.integrate; imports, file access, arbitrary attributes, and ^ powers are blocked. Click Validate before saving when you want an explicit check.

Custom models appear in Curve Fit as Custom: model name and are stored inside the .jasc project.

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12. Plotting and the Plot Inspector

The Plot tab holds one editable plot draft. A plot can contain multiple queued series, including series whose X, Y, and Z values come from different datasets.

Build and edit a plot

1. Open the Plot tab. Click New Plot first if you want to replace the current draft; JASC asks what to do with unsaved work.
2. In Series, choose a plot family and subtype.
3. Configure the data mappings and type-specific appearance.
4. Click Add Series. The series is queued and the plot is rendered.
5. To add another series, click New Series, adjust the retained mappings and appearance, and click Add Series again.

Selecting a queued series switches the editor to Update Series mode. Mapping, subtype, and appearance edits are staged: they do not alter the queue or canvas, even when Live preview is enabled. The status line identifies unapplied edits. Click Update Series to commit them or Revert Edits to reload the queued values. Until you choose one, JASC keeps that series selected rather than silently losing the staged edits, and it will not save the plot. Use Remove Selected or Clear All to edit the queue. Right-click the queue for New, Update, Revert, Duplicate, Remove, and Clear actions.

Edit directly on the plot

Rendered data, text, and axes can be edited without navigating to the lateral inspector. Double-click:

- A plotted line, marker, bar, filled region, distribution, or composition cell to edit that committed series. The focused dialog provides compatible graph subtypes, series name and legend inclusion, continuous/dashed/dotted/dash-dot line patterns, line width, symbol shape and size, automatic or custom colour, fill opacity, and bar width. For example, an XY line can be changed directly to scatter, line + markers, errorbar, bar, horizontal bar, or area.
- A 2D or spectral map to open its map-specific Colour & Grid, Contours, Validity, and Extrema editor.
- The plot title to change its text.
- An X, left-Y, or right-Y axis label, tick-label area, or axis line to edit its label, label size, tick-number size, linear/log scale, optional minimum/maximum, and the shared grid state.
- A legend entry label to rename that committed series or include/exclude it from the legend.
- The legend title, frame, handle area, or other empty legend space to edit visibility, position, text/title sizes, title, columns, frame and opacity, marker scale, handle length, and locator/confidence-band inclusion.

Accepted changes redraw immediately and update the corresponding lateral Plot Inspector controls; the inspector remains available for the same settings. Direct series and legend-entry edits respect staged-series protection: use Update Series or Revert Edits first if the selected series has unapplied changes. Hovering over the canvas also shows a reminder about the available double-click targets.

Plot types and mappings

Family	Subtype	Data mapping and special controls
XY / Categorical	Line	X and Y; line style/width; optional left/right Y axis
XY / Categorical	Scatter	X and Y; marker and size; optional left/right Y axis
XY / Categorical	Line + Markers	X and Y; line and marker controls; optional left/right Y axis
XY / Categorical	Errorbar	X/Y with symmetric errors or separate lower/upper percentile endpoints
XY / Categorical	Vertical Bar	Numeric positions or text categories plus values; bar width and fill alpha
XY / Categorical	Horizontal Bar	Numeric positions or text categories plus values; bar width and fill alpha
XY / Categorical	Area	X/Y filled area, line controls, fill alpha
Distribution	Histogram	One value column, 1-1,000 bins, density, cumulative, fill alpha
Distribution	Box Plot	One or more value columns; mean, median, outlier controls
Distribution	Violin Plot	One or more value columns; mean, median, extrema, statistic-line width
Composition	Pie	Label column plus positive value column
Composition	Donut	Label column plus positive value column
Composition	Voronoi Composition	Label/value columns and Circular, Rectangular, or Polygonal container
2D Maps	Interpolated Heatmap	X/Y/Z coordinates; native complete grid or interpolated scattered grid
2D Maps	Contour	X/Y/Z coordinates; configurable levels, labels, line style, and normalization
2D Maps	Filled Contour	X/Y/Z coordinates; configurable fill levels and optional contour overlay
Spectra Plot	Spectral Heatmap	Long-form X/Y/Raman-shift-or-frequency/intensity data; target center; closest or linear spectral sampling

Spectra Plot	Spectral Contour	Long-form spectral-map data rendered as contour lines at the target center
Spectra Plot	Filled Spectral Contour	Long-form spectral-map data rendered as filled contours at the target center

2D-map interpolation choices are Linear, Cubic spline, and Piecewise cubic. Automatic rendering preserves a complete Cartesian input grid without resampling and otherwise interpolates onto the selected resolution; users may explicitly require Native or Interpolated mode. The Palette selector applies the supplied Matplotlib colour maps and is retained in saved plots. Colour-bar label sets a custom intensity/Z label; leaving it blank uses the selected value column automatically. The rendered colour-bar label can also be edited directly by double-clicking it.

Select a committed map and click Map Details..., or double-click the map, for four groups of saved controls:

- Colour & Grid: automatic, manual, or robust-percentile limits; linear, logarithmic, symmetric-log, or center-aware diverging normalization; missing/under/over colours; opacity; and automatic or equal-unit aspect.
- Contours: level count, fixed interval, or explicit numeric levels; labels and number format; line colour/width/style; and an optional line overlay on filled contours.
- Validity: measured data points, Delaunay triangulation, convex-hull boundary, and masking outside that measured boundary.
- Extrema: global and local maxima/minima, prominence and physical-separation filters, label count/content, and either measured or rendered-grid values. Labels explicitly state measured or interpolated so interpolation overshoot is not presented as an observation.

Click Analyse Map... to create an ordinary reusable worksheet containing the rendered X/Y/Z grid, a closest horizontal or vertical profile, an arbitrary point-to-point profile, rectangular ROI statistics, or contour vertices and enclosed areas. ROI output includes cell count, mean, median, standard deviation, extrema with coordinates, and a grid-spacing-weighted integral. The repository includes analytically defined regular-grid and irregular point-cloud examples under Test_Data.

Horizontal, vertical, and arbitrary line cuts also create named, editable plots under Workspace > Plots, backed by the generated profile worksheet. They are saved with the project. The source map stays active, so you can create several cuts before opening their plots. Use Pick Cut Line... to click the start and end directly on an ordinary or spectral map, then confirm the endpoints and sample count.

Peak/Raman result maps provide Map Cuts... for horizontal, vertical, or arbitrary cuts of the currently selected parameter or quality metric, and Pick Cut Line... for mouse-picked endpoints. These cuts respect Hide failed pixels and publish only the cut worksheet and saved plot, without exporting the complete internal result tables. Right-click or press Esc to cancel picking; changing the metric or analysis also cancels it. Arbitrary-line values are spatially interpolated, rather than additional measured pixels.

A dedicated row above the canvas follows the mouse and reports x, y, and interpolated Z/intensity for ordinary and spectral maps; masked interpolation edges fall back to the closest measured map pixel. Maps require numeric Z values and at least three distinct non-collinear X/Y coordinate points. Rendering failures appear as a message on the canvas instead of a blank plot.

For an imported hyperspectral map, choose Spectra Plot and select its worksheet. JASC automatically proposes X, Y, Raman Shift (or a frequency/wavenumber equivalent), and Intensity. Enter the target center and choose Closest measured shift to take the nearest available spectral sample independently at every pixel, or Linear spectral interpolation to interpolate between the two surrounding samples in every pixel spectrum. The separate 2D-map interpolation control then determines how those sampled XY intensities are drawn on the spatial grid. Target center, spectral column, sampling method, spatial interpolation, and grid resolution are retained in saved plots.

Composition plots cannot be mixed with XY series, and only one composition series can render at a time. Pie/donut require positive values. Voronoi regions are value-weighted approximations and automatically reduce label density for large category counts.

Common series appearance

Depending on plot type, the inspector exposes:

- Custom legend label and per-series Show this series in legend.
- Solid, dashed, dotted, dash-dot, or no line.
- Circle, square, triangle, diamond, cross, or dot markers.
- Automatic style colours, 20 preset colours with swatches, or a custom RGB/hex colour.
- Line width, marker size, fill alpha, and bar width.
- Left or right Y-axis assignment.

Column roles preselect likely mappings. Axis-label fields prefill from Long Name and Unit metadata while they remain blank or automatically managed; manually typed labels are not overwritten.

In Series > Color, choose a preset or click Custom... to open the full colour picker. Select visually, enter RGB values, or enter an HTML hex value such as #13a2a7. Use Update Series to apply the choice; the series colour controls its lines, symbols and supported fills. Auto (style cycle) restores the active style's automatic colour. Custom choices remain available in that editor's dropdown and are retained with applied plots, pending tab edits, project saves and Python recordings. Data Location > Color offers the same custom picker for locators. Double-clicking a plotted series also opens its focused colour/appearance editor.

Axes, style, and figure

The Axes tab controls:

-

Title; X, left-Y, and right-Y labels.
- Per-axis label and tick-number sizes for X, left Y, and right Y. Style default leaves typography under the selected Matplotlib/SciencePlots style; any positive point size overrides it on the live canvas and exported output.
- Linear or log scale independently on X, left Y, and right Y.
- Optional minimum/maximum for each axis.
- Grid visibility.
- Export size basis plus width (3-20 inches) and height (2-15 inches). Choose Full exported image to size the complete output, or Plotting area (axes box) to size only the inner rectangle bounded by the X and Y axes.
- Style preset: Matplotlib default, Science, IEEE, Nature, or Notebook.
- Color cycle: Default, Bright, or High visibility.
- LaTeX text on/off.
- Custom comma-separated SciencePlots style tokens.

Log ranges must be positive. Invalid or reversed ranges mark the preview out of date without interrupting typing with modal dialogs. SciencePlots styles require the dependency in requirements.txt; LaTeX rendering also depends on a working external LaTeX installation.

The interactive plot always expands to the available canvas. Applying a style or changing export width, height, or size basis does not resize the on-screen canvas. Physical dimensions are applied only by the toolbar export action. In plotting-area mode, JASC calculates the larger complete figure required to keep the requested axes-box dimensions while retaining titles, tick labels, margins, color bars, and outside legends.

Legend

The Legend tab controls:

- Visibility.
- Automatic or standard inside positions.
- Outside right, outside top, or outside bottom placement with reserved export space.
- Text size, optional title, and title size.
- Column count.
- Frame visibility and opacity.
- Marker scale and handle length.
- Whether locator markers and confidence bands appear.

Entries follow queued-series order across both Y axes, and duplicate labels are removed.

Live preview and Refresh Plot

Live preview is off by default. Axes, Legend, and style changes mark the canvas out of date; click Refresh Plot when ready. Enabling Live preview redraws those plot-wide changes automatically after a short debounce. Export-size changes are stored without redrawing because they affect saved files only. Series edits always remain staged until Add Series or Update Series is clicked, which avoids accidental queue changes while experimenting with mappings or appearance. The preview indicator reports current, pending, out-of-date, invalid, or failed state.

Data locators

Open Locator, select a queued series, and choose a lookup:

- Find Y from X interpolates Y on an ordinary XY series.
- Find X from Y finds every piecewise-linear crossing on an ordinary XY series.
- Surface plots use entered X and Y to interpolate Z.

Choose marker shape, size, and color, then click Add Locator. Located values appear in the list and on the correct left/right axis. Right-click the located-items list to add, remove, or clear locators. Non-monotonic curves can return several X crossings; repeated shared vertices are deduplicated. A horizontal segment at the requested Y is reported as ambiguous instead of returning an arbitrary X.

Matplotlib canvas toolbar

Use the embedded Matplotlib toolbar for home/reset, back/forward, pan, zoom, subplot configuration where offered, copying, and saving the rendered figure. Copy places the current on-screen canvas image on the system clipboard without applying export dimensions. The same command is available as the top-row Copy Plot button and as Plot > Copy Current Plot to Clipboard (Ctrl+Alt+C). The save action applies the configured full-image or plotting-area physical size only for export, then restores the responsive canvas. Both operations produce static images; neither saves editable JASC plot state.

Independent plot documents and tabs

Plots contains a tab for each open plot. New Plot and + create another independent project document (automatically named Plot 1, Plot 2, etc.). The Workspace list includes both open and closed plots. Opening one that is already open activates its existing tab and keeps its edits.

- Edit each plot through the shared inspector, which follows the active tab.

Series, axes, legend, locators, staged fields, viewport, and edit history belong to that plot. Undo/Redo act on the active plot while Plots is selected.

- Analysis-generated plots open as new documents. To combine results, open the target plot and add another series from the result worksheet.
- Close a tab with x to retain its document in the workspace. Reopen it from

the list to restore its pending fields and view. Closed canvases are released.

- Rename changes a document's name without changing its identity. A name already used by another plot is rejected. Duplicate Plot creates an independent copy with a new identity.
- Delete Plot removes the document after confirmation. It is separate from closing a tab and does not delete its source worksheets.

Use Pop Out on an individual plot or comparison to move its live view and inspector into a separate, maximizable window. Its tab keeps a Return to Tab notice. Close the window or use Return to Tab to reattach it without reloading the plot, losing pending fields, or resetting its zoom. Activating a plot window selects the corresponding document for plot actions. Project save includes edits in these windows; project/app closure still protects unsaved changes.

Save Project (Ctrl+S) writes all current plot documents, including closed plots and pending editor fields. There is no separate Save Plot step. A tab and project asterisk indicates plot changes since the last successful project save. Use Update Series to apply staged series changes to the drawing; saving a project preserves staged fields without silently applying them. Existing project files without document metadata still open, and signing/encryption remain available for the complete project.

Closing the application or replacing/closing a project with unsaved plot changes offers Save, Discard, or Cancel once. Cancelling the file/password/key prompt or a failed save keeps the project open. Plot edit history is retained within the session, including tab close/reopen; it is not stored across project reloads. Source-structure changes clear affected history to avoid restoring old bindings.

Comparing plots in an equal-cell grid

Choose Plot > Compare Plots... (also available in the Workspace plot context menu) once at least two plots contain applied series. Check 2-16 source plots, drag them or use Move Up/Down to set panel order, and enter a comparison name. Choose 1-4 rows and columns or use Fit Grid. Every cell has equal width and height, including empty trailing cells. Show panel labels adds A, B, C, etc.

The default result is a new comparison tab in Plots. Enable **Open in a separate window in the picker or use Pop Out** later. Its controls move with the comparison. Return to Tab or closing the separate window reattaches it; closing its document tab releases the view and retains the layout in Workspace. Save Project persists the comparison's sources, order, grid, matching options, and viewport. Comparison layouts support the usual per-document Undo/Redo.

Use Edit Comparison... or Choose / Reorder Plots... to revise the selection, and the inspector to change rows, columns, matching, or panel letters. Comparisons are native renderings of the source definitions. They retain plot types, styling, legends, secondary axes, colour bars, and aspect ratios without changing source plots. Equal cells do not force identical inner axes when legends, colour bars, or fixed aspect ratios require space. Existing comparisons cannot be nested.

Synchronize X axis and Synchronize Y axis use a common numeric range and link subsequent wheel, pan, and zoom changes within the comparison, including heatmaps and contour images. Y synchronization uses the left axis.
Matching is skipped, with a message, for missing/unrenderable sources, categories, composition/distribution panels, or different axis scales. Units are not converted; choose matching only for comparable quantities. Pan, Zoom, and **Reset View** control the comparison without changing original viewports.

Source plot edits refresh after a short debounce; Refresh also rereads worksheet values. Pending source series fields must be applied with Update Series first. Double-click a panel to open its source plot. Renaming a source keeps the link; deleting it leaves a visible placeholder that can be replaced through selection.

Copy captures the full grid, excluding controls. Export PNG... renders 1000 x 750 pixels per cell into one PNG, independently of window size, then restores the live view. Vector/PDF comparison export, shared legends, inset panels, and a general page composition editor are follow-ups.

Plotting asymmetric bounds

Choose the Errorbar subtype and enable the X/Y percentile controls in the Series inspector. Select endpoint columns and probability levels, then Add Series or Update Series. Each axis may independently use symmetric errors or percentile bounds. The endpoint columns belong to that axis's selected worksheet.

Bounds must bracket the displayed value. Zero-width sides may be drawn; fitting requires positive widths. Logarithmic axes require positive endpoints. Probability levels, centre/origin meaning, applied mappings, and pending edits persist with the project and follow column renames. Closing/reopening a tab or arranging comparison panels preserves the unequal error bars. Plotting mean/mode/other estimates or posterior quantiles does not turn them into an eligible measurement likelihood.

Navigating with the mouse

- Wheel up/down zooms in/out around the pointer.
- Shift+wheel pans by a fraction of the visible range.

- Middle-button drag pans continuously; release, leave the canvas, or press Escape to end the drag.

The pointer's position at the start of a gesture chooses its axes. Over the X spine, ticks, or label, only X changes. Over a Y spine, ticks, or label, only that Y axis changes, including the right axis when present. Inside the plot, X and Y change together, including both left/right Y axes for overlays. Dragging keeps its original axis selection even when the pointer moves outside the plot area.

These gestures work in tabs and popped-out windows, including linear/logarithmic and reversed axes, and participate in Matplotlib Back, Forward, and Home. They leave source data and applied plot definitions unchanged. On image comparisons, enable the X/Y synchronization options to navigate matching spatial regions together; turn them off to inspect panels independently. Units and colour scales are not converted or synchronized by these spatial-axis options.

Map colour bars do not trigger spatial navigation. Wheel events handled by a plot do not also scroll the surrounding comparison. Navigation is suspended during map-cut picking or an active toolbar selection/drag so the tools do not compete. Paused trackpad wheel input commits one navigation-history step per burst.

---

13. Batch Processing

Batch-capable workbenches are:

- Apply Formula.
- Derivative and Integral.
- Smoothing.
- FFT and Inverse FFT.
- Interpolation.
- Statistics.
- Curve Fit.

Standard batch workflow

1. Configure the operation using the current sheet as the reference.
2. Select one or more entries in Apply to sheets.
3. Select Exact name or Column position matching.
4. Configure output names/conflict behavior where offered.
5. Confirm the operation dialog.
6. Review Batch Preflight Preview, including resolved inputs, outputs, status, and error details.
7. Click Run Batch.
8. Monitor the cancelable progress dialog.

Preflight rows marked failed or blocked explain incompatible columns, invalid mappings, or output conflicts before execution. Compatible jobs can still run when other rows fail, depending on the operation's plan.

Column binding

- Exact name resolves each requested header literally on every target sheet. Use it for standardized worksheets.
- Column position treats the reference mapping as col1, col2, and so forth and resolves the same positions on every target sheet. Use it for repeated exports whose headers differ but layouts match.

Column roles influence initial dialog choices but do not replace batch binding.

Output naming and collision policies

Where the dialog supports them:

- {source} expands to the source sheet name.
- {column} expands to the processed signal/value column.
- Create unique protects existing columns by generating a noncolliding name.

- Stop and warn blocks the colliding output.
- Overwrite/update permits replacement after the preflight makes the target explicit.

Dataset-producing operations also create unique names when a requested result name already exists.

Multiple input columns

- Derivative, Integral, Smoothing, FFT, Statistics, and Curve Fit can process multiple selected signal/value columns.
- Inverse FFT accepts zero, one shared, or one-per-real imaginary column.
- Curve Fit accepts zero, one shared, or one-per-Y sigma/weight column.
- Pair statistics accept one shared secondary column or one-per-primary secondary columns.
- Interpolation accepts multiple value columns.

Parallel formula execution

Apply Formula uniquely exposes Auto, Sequential, and Parallel processes plus maximum workers. Auto decides whether parallel processing is worthwhile for the headless target sheets. The visible current sheet may be handled safely in the GUI path while compatible background jobs use worker processes.

Progress, cancellation, and summaries

Peak/Raman batch/map fitting and baseline-only map correction use status-bar progress and Cancel controls described above; cancellation publishes no partial results.
Other batch operations display a modal progress dialog and check for cancellation between jobs/results. Their cancellation does not undo jobs already completed.

Multi-job runs create operation-specific summary datasets such as:

- _formula_batch_summary
- _derivative_batch_summary / _integral_batch_summary
- _smoothing_batch_summary
- _fft_batch_summary / _ifft_batch_summary
- _interpolation_batch_summary
- _statistics_batch_summary and sometimes _statistics_batch_wide_summary
- _fit_batch_summary

Summaries include source sheets/columns, resolved mappings, requested outputs, status/errors, execution details, and operation-specific metrics.

Record Python API calls

Record Macro captures successful application operations as chronological Python calls to the public jasc_api.Session API. The resulting .py file can be inspected, edited, and versioned as normal Python source. Plot creation and later edits appear in order as separate calls.

While recording, the main window displays a red edge and a persistent Recording (.py) badge with a Stop Recording button in the status bar. The edge follows window resizing and adapts to light/dark palettes. Other status messages do not hide the badge. Both indicators disappear when recording stops, including when stopping to save the Python script.

1. Save a starting project if the recording will use existing workspace sheets.
2. Choose Macros > Record Macro, then perform operations normally. File imports retain their paths and parser settings; existing sheets become explicit session.sheet(...) references.
3. Open View Recorded API Calls to inspect the script while recording.
4. Choose Stop Recording > Save Python API Script and edit the saved file in your Python editor if desired.
5. Choose Macros > Run Python API Script, open the file, review its source, choose the permitted input folder and output parent, and click Run Recording. Playback uses the current workspace as its starting state. Alternatively, from the repository root, run it with the project interpreter:

```
python -m jasc_api recording.py --project starting.jasc --output results.jasc
```

Use the project saved before the recorded transformations as the input. --project is optional for scripts that import their own data; --output saves the final workspace as a new project. Add --input-dir to select the only folder from which a recording may import files; the default is the recording's folder. --output-dir selects the parent of the generated run folder. Recorded exports execute in their original order inside staging, then become visible together after success. The JSON batch-recipe dialog remains a separate workflow.

The file has this general form (illustrative parameters):

```
from jasc_api import Session, series

def run(session):
    sheet = session.sheet("sample")
    session.smooth(sheet, x_column="x", y_column="y", method="savgol",
                   output_column="smoothed", window_length=11)
    fit = session.fit(sheet, x_column="x", y_column="smoothed",
                      model_key="linear", method="trf")
    plot = session.plot([
        series(x=(sheet, "x"), y=(sheet, "smoothed")),
        series(x=(fit.curve, "fit_x"), y=(fit.curve, "fit_y")),
    ], name="Fit")
    session.export_plot(plot, "fit.png")
```

GUI recording now covers the following operations:

Area	Recorded calls and behavior
Imports	Text-file parser choices, Excel worksheet tabs/header choices, and disk-backed large-data imports.
Worksheets	New, duplicate, rename, delete, cell edits, paste, row/column insert/delete, resizing, clearing, Fill Range, Smart Fill, column metadata/renaming, and undo/redo.
Data Prep	Filter, positional subset, sort, duplicates, missing values, binning, grouping/aggregation, joins, and reshape.
Numerical analysis	Smoothing, derivative, integral, sequential/parallel Formula results, FFT/iFFT, all existing statistics scopes, and interpolation using generated, manual, or existing-column query points.
Fitting	Curve Fit with seeds, bounds, fixed states, weights, percentile uncertainties, residuals, confidence settings, and custom-model definitions.
Spectral analysis	Single-spectrum baseline/detection/composite fitting, explicit reviewed peak decisions, baseline maps, Peak/Raman batch/map fits, analysis duplicate/delete, and exporting internal analysis tables as worksheets.
Maps and plots	Grid, axis/line cuts, ROI summaries, contour extraction, applied plot definitions/style updates/renaming/deletion, and comparisons.
Results and files	Recomputed batch summaries, wide statistics tables, parameter correlations, fit reports, large-data window extraction, data/figure export, and ordinary project saves.

Results refer to earlier sheet, fit, and analysis variables. Batch tables are recomputed from these results at the corresponding step, including changed statistics group counts; measured values are not embedded as frozen result tables. Original GUI timings are replaced with zero in replay summaries. Failed or cancelled audit rows describe the original run; the script executes recorded

successful operations and stops if a replayed operation raises an error. Background spectral calls are captured only after successful publication; cancellation, stale inputs, and failed publication do not add analysis calls.

Reviewed peak enable/disable, removal, manual addition, and repositioning choices are retained. Replay detects fresh candidates and matches explicit review edits within the recorded physical-axis tolerance; an unmatched reviewed peak raises an error. Baseline and fit calculations use the replay input data. Template fit seeds remain independently reused for each fit.

The API also provides load_folder() so authors can write ordinary Python loops over folder files and combine them with sheet() references to existing data:

```
for sheet in session.load_folder("measurements"):
    clean = session.prepare(sheet, operation="missing", name=sheet.name+"_clean",
                            columns=["signal"], strategy="fill_mean")
    report = session.statistics(clean, operation_key="single_summary",
                                primary_column="signal", name=sheet.name+"_stats")
```

Image-digitizer extraction, project replacement during a recording, internal spectral-viewer presentation changes, and running another macro or batch recipe inside a recording are not yet supported by recording/replay adapters. Numerical spectral analysis and regular plot/comparison styling are supported as listed above. For images, digitize manually first, then apply macros to the extracted worksheets or exported CSVs. Future image replay would need explicit calibration, axis scales, regions, and trace-selection settings, with human review for ambiguous images; this is an adapter/design gap rather than a blanket security ban on image data. Encrypted/signed saves and security-key management are excluded from recordings and restricted playback; passwords and private keys are never written to scripts. Unsupported data changes, missing dependencies, or unknown outputs make a recording explicitly incomplete, with an explanatory error before its calls execute. Save a starting project before recording, inspect the script, and use Save Python API Script for this broader coverage; Review / Run Batch Recipe continues to support its smaller set of recipe steps. Numerical operations run without a window; map extraction and figure export lazily use the Qt renderer.

Restricted playback and Windows executables

JASC reads recordings as a restricted command language; it does not import the file or execute it through exec, eval, compile, or runpy. The complete outer syntax is validated before any command runs. Approved session calls, literal values, named result references, the series helper, recorded count helpers and bounded loops over worksheet collections are supported. Imports other than the recognized JASC declaration, decorators, arbitrary attribute/method

access, assignment to API objects, comprehensions, general control flow, expanded arguments, shell commands and arbitrary built-ins are rejected. Mathematical formula/model strings retain the existing numerical expression allowlists plus playback complexity/allocation checks; they cannot access general Python objects.

The review dialog shows the exact captured source. Changing the file after review does not change the bytes that will run. Existing workspace data is copied to a starting snapshot. A separate worker process performs numerical operations and rendering, with a default two-minute time limit and 2 GiB memory limit. The GUI remains available; Cancel terminates the child and discards staged changes. Native calls in the child need not cooperate with Python thread cancellation. Preparing or loading the starting/result project may finish its current work unit before cancellation takes effect. Closing JASC also waits for the worker to stop. A changed worksheet, metadata, plot or project prevents publication, so replay cannot silently replace intervening user work.

File imports are limited to the selected input folder and approved data formats. Traversal, URLs, UNC/device paths, alternate data streams, ambiguous/reserved Windows names, symbolic links and directory junctions below the access root are rejected. Relative exports retain their subfolders in a new output run folder; recorded absolute destinations use their filename there. Only data, ordinary .jasc projects and supported figure formats may be exported. Existing outputs, including duplicate exports within one run, cannot be overwritten. _jasc_result.jasc contains the completed workspace and _jasc_run.json records the script hash and run information. Encrypted/signed saves and security-key operations are excluded. Style files/URLs/module paths and external LaTeX tools are also excluded; installed Matplotlib/SciencePlots style names remain available without external TeX.

These controls restrict JASC recordings; they are not an operating-system sandbox for arbitrary Python/native libraries. Review the approved operations and grant only the input folder needed for the task. Do not run downloaded .py files with the Python interpreter itself: top-level Python code can execute before any JASC launcher is reached. New generated files delegate their normal launcher to the restricted runner, while older generated launchers remain readable by that runner.

Build Windows distributions using python build_windows.py after installing requirements-build.txt in the build environment. JASC.spec produces a one-folder bundle with JASC.exe for the GUI and JASC-cli.exe for console output. Both accept:

```
JASC-cli.exe --run-script recording.py --project starting.jasc --input-dir measurements --output results.jasc
```

Distribute the entire generated JASC folder, including _internal. Python and the numerical/Qt runtime are bundled, so recipients do not need to install Python. --self-test constructs and closes the GUI offscreen for packaging checks.

Automated bundle tests remove Python from PATH, set unusable Python environment paths, exercise both executable entry points, perform a fit/figure export and verify that injected Python does not create a marker file. This is distinct from validation on a fresh Windows VM, which remains a release check. Installer, code-signing and cross-platform distributions remain separate packaging work.

During development, regenerate the PyInstaller executable package only when the user explicitly requests a build or rebuild. Ordinary implementation, fixes, documentation, and source testing do not trigger packaging. An existing bundle contains a snapshot of the application and its manual; updating source files or regenerating this PDF does not refresh an existing executable distribution.

Runnable synthetic macro example

Test_Data/Macro_Example contains three supplied CSV spectra, the restricted process_spectra.py macro, independent expected_truth.json, a reproducible data generator, and comparison_preview.png. Detailed GUI, Python CLI, and packaged Windows commands are in Test_Data/Macro_Example/README.txt.

1. Start a new, empty project so the example's sheet and plot names are available.
2. Choose Macros > Run Python API Script and select Test_Data/Macro_Example/process_spectra.py.
3. Set the allowed input folder to Test_Data/Macro_Example, not its input subfolder. The macro imports the relative input folder below that access root.
4. Choose an output folder, for example Test_Data/Macro_Example/results, and click Run Recording. Each run writes into a fresh subfolder.
5. Inspect the processed sheets, fitted parameters, individual plots, and the synchronized three-panel comparison. Start another empty project before retrying.

The macro removes missing/duplicate samples, sorts X, subtracts an exact synthetic reference background, smooths for display with LOWESS, and fits the unsmoothed corrected observations using their known measurement sigmas. Each input has 407 rows and each processed sheet has 396. A successful run produces 18 worksheets, four plots, 15 CSV exports, and four PNG exports plus the result project and run manifest. The fit does not read the independent truth file. Its 95% bands describe uncertainty of the fitted mean curve, not prediction intervals for observations. The optional generate_data.py is a trusted developer utility, not a restricted macro; the supplied inputs are ready to use without running it.

Batch recipes for folder/workspace replay

The Macros menu records a sequence of supported operations and ordinary XY plot definitions, then repeats it for selected workspace sheets, disk-folder

files, or both. This first release supports Smoothing, Curve Fit, and line/scatter/line+marker/errorbar plots. Text-file import settings can be captured while recording; folder inputs currently support CSV, DAT, and TXT.

1. Choose Macros > Record Macro and enter a name. Use the selected worksheet as the reference, or import representative data before applying operations.
2. Demonstrate each operation on one sheet and one Y column using the normal Smoothing and Curve Fit dialogs. A later fit may select the smoothed column. Batch targets are chosen when replaying rather than during the demonstration.
3. Create and style ordinary XY plots as usual. Stop Recording collects the final applied definitions of plots created or changed during the recording.
4. Choose Review / Run Batch Recipe. Inspect the ordered steps; edit supported settings, remove steps, or move independent steps. Dependencies prevent removing/reordering a step that later results or plots require.
5. Check workspace sheets and/or use Add Files or Add Folder. Include subfolders when needed. Imported columns match the reference by exact name; there is no implicit positional or semantic-role fallback in this release.
6. Select Preview Compatibility. Review ready/skipped inputs and their reasons. The preview checks structure and source numeric metadata; numerical requirements and optimizer convergence are also checked during execution.
7. Optionally choose Export Tables To..., then Run Macro. Computation runs in the background with progress and Cancel. The first generated plot can open automatically; remaining plots are available in Workspace > Plots.

Each successful folder input becomes a normal source worksheet. Replay of an existing worksheet leaves it unchanged. Derived results, fitted curves, residuals, and plots are separate project objects with collision-safe names. The run summary records source identity, operation/step, status, timing, warnings, and fit parameters/errors and metrics. Its worksheet metadata retains the macro definition; generated fit worksheets retain the full fit report.

Only complete input results are published. A failed input adds a failure row while other inputs continue. Cancel retains already completed inputs and discards the incomplete input. Changed/replaced workspace sources are rejected before publication; changed folder files require another compatibility preview. Native numerical calls may return before cancellation takes effect. Source snapshots and retained outputs use memory; this is not an out-of-core runner.

Save Batch Recipe writes a standalone .jasc-macro.json file. Load Batch Recipe opens it for review and reuse with different inputs. Save the project separately to retain the generated worksheets/plots. Unsaved macro files and macro-generated project outputs have save/discard/cancel protection. Optional CSV export writes tables, the summary, and macro definition into a new jasc_macro_run_... directory; existing files are not overwritten, and these generated run directories are excluded from recursive folder input discovery. Figure export remains available

through the existing plot export commands.

Unrecorded changes to a reference/result sheet, unknown dependencies, and unsupported plot kinds produce recording gaps that block replay. Start a clean recording from the intended input to resolve them. The recorder does not capture arbitrary cell editing, every analysis family, comparison creation, or export commands. Fit seeds/constraints repeat independently on each input. A transformed Y signal cannot reuse raw measurement sigma/percentile bounds without an explicit propagation rule; such weighted chains are blocked rather than inferred.

14. Export, Persistence, and Large Data

Export a dataset

Select a dataset and choose File > Export Dataset... (Ctrl+E). Choose CSV or tab-delimited text. JASC exports column headers and values without the pandas row index.

Dataset export does not include formulas, column metadata, saved plots, or custom models. Save a .jasc project when those need to be preserved.

Export a plot

Use the save button in the Matplotlib toolbar. The available raster/vector formats depend on Matplotlib. Outside legends reserve figure space in the exported output. To reuse the current screen rendering quickly, use Copy Plot, the toolbar Copy action, or Ctrl+Alt+C; clipboard copy uses the live canvas size, while file export uses the configured physical dimensions.

An exported image is static. Save the plot definition and project when later editing is required.

Large-data thresholds and behavior

- Source files of at least 100 MiB trigger a pre-import warning.
- A worksheet is flagged as large at 2,000,000 cells or an estimated 250 MiB of DataFrame memory.
- Column sizing uses at most 100 sampled rows normally and 25 rows above the 250,000-cell responsiveness threshold.
- Clipboard parsing, worksheet growth, range fill, and rectangular assignment use block-oriented operations. Right-click menu availability checks inspect compact Qt selection ranges rather than materializing every selected cell, so selecting a large table does not make menu opening proportional to its cell count.
- .jasc worksheet CSV entries stream from the ZIP archive during project load.
- Undo/redo retains at most 20 snapshots and approximately 256 MiB per stack.

Ordinary editable worksheets remain in Pandas memory. Text inspection/parsing runs in the background; Excel and project I/O still run synchronously. Autosave and recovery are not implemented.

Open a disk-backed large dataset

Choose File > Open Large Dataset... or Workspace > Large Data > Open Large Dataset. Select a regular comma-, semicolon-, tab-, or pipe-separated .csv, .dat, or .txt table, or an existing .parquet file. Text files use the usual import preflight and parser options. Large-data mode currently rejects whitespace-separated tables and vendor-specific spectrum/map layouts; use ordinary Import Data for these.

JASC creates a private Parquet snapshot on disk through Polars rather than retaining a complete source DataFrame. Text values remain strings, including numeric-looking values, identifiers with leading zeros, and late mixed tokens. Number-format choices are retained as provenance; this mode does not perform whole-column numeric inference. Existing Parquet types are retained. The original file is not modified, and later changes to it do not affect the snapshot.

The central Large Data tab is read-only and shows one window of at most 1,000 rows and 32 columns, with an 8 MiB data-buffer limit. Oversized windows shrink; a row exceeding the limit produces a controlled error. Column widths are fixed initially. This budget covers the displayed data, not all process memory: workers, Parquet decoders, Qt, and import buffers require additional memory.

- Scroll past the bottom/top edge to load adjacent rows, or use Previous rows / Next rows.
- Enter a global, one-based row number and choose Go to row to jump directly.
- Use Previous columns / Next columns to browse wide tables in bounded column windows.
- The summary and row labels retain positions in the complete source. Copy (Ctrl+C) copies only selected cells in the current window.
- Reads run in a worker. Old windows are evicted, rapid requests coalesce, and stale results cannot replace a newer source/window. Retry is available after a read failure.
- Create worksheet from this window explicitly copies the displayed rows/columns into an ordinary editable worksheet with source-range provenance. This is a subset; analysis on that worksheet covers that subset. Text-import values remain strings; ordinary numeric-string validation is available in the analysis workflows.

Large datasets appear separately from ordinary worksheets, so existing formulas, plotting, batch analysis, and snapshot undo cannot silently operate on a partial source. Full-source filtering, sorting, aggregation controls, reduced plots, and bounded spectrum extraction remain future work; the backend exposes a lazy full-source query boundary for those operations.

Save the project to retain large sources. Projects containing them use version 9 and embed Parquet members; load streams those members back to private disk snapshots. Original source files are not needed to reopen the project. Ordinary-only projects continue to use version 8. Package seals, signatures, and encryption cover the embedded data as usual. Project save/load remains synchronous, so very large packages can pause the interface despite bounded payload memory.

Opening requires temporary disk space for the snapshot and, for text, an intermediate UTF-8 copy. Snapshots are unencrypted files in the operating system's temporary directory while open, including sources loaded from encrypted projects. Removing a source releases its snapshot once any active reader finishes. Import Cancel checks streaming-copy boundaries and rejects pending results; native Polars conversion cancels after that phase finishes. Closing waits safely for active workers.

---

15. Troubleshooting and Current Limitations

JASC does not start

Run from the repository root and verify the active interpreter:

```
python --version
python -m pip install -r requirements.txt
python main.py
```

If python is not found, activate the environment where the dependencies were installed or invoke that environment's Python executable directly.

A CSV/DAT/TXT file parses incorrectly

- Review the preflight row, especially detected metadata rows, header status, and preview values.
- Auto detects comma, semicolon, tab, pipe, and whitespace. If an unusual or inconsistent file is parsed incorrectly, reimport and choose an explicit delimiter.
- Force First tabular row contains column names or No header row when header inference is wrong.
- Choose a fixed number of leading rows when an instrument preamble cannot be distinguished reliably from the data table.
- Save the source as UTF-8.
- JASC does not currently choose encodings or provide per-file manual overrides inside one batch; split unusually inconsistent files into separate imports when they require different forced rules.

A numeric operation rejects a column

- Open Column Metadata and confirm it is Auto, Numeric, or Scientific rather than Text.
- Remove or correct non-numeric text in explicitly Numeric/Scientific columns.
- Check for duplicate X values, missing aligned points, or uneven spacing where the selected method requires uniform samples.

A formula fails

- Use **, not ^, for powers.
- Use col("Exact Header") for spaces, punctuation, keywords, or names that conflict with functions.
- Check the Apply Formula Help panel for the curated function list.
- Ensure a vector expression returns exactly one value per worksheet row.

A fit does not converge

- Use realistic initial guesses.
- Confirm lower bounds are smaller than upper bounds and do not exclude the solution.
- Try Trust Region Reflective or Dogbox for bounded fits.
- Inspect missing, nonfinite, or nonpositive uncertainty values.
- Reduce the number of free parameters or fix known parameters.
- Confirm the model domain is valid for the selected X range, especially log-normal, power-law, Bessel Y, or fractional-power models.

A 2D map is blank or reports an error

- Choose distinct X, Y, and Z mappings.
- Provide at least three distinct, non-collinear coordinate points.
- Ensure Z is numeric and enough finite points remain after blank removal.
- Try Linear before cubic methods and reduce grid resolution for expensive data.
- Use Automatic or Interpolated grid when the X/Y points do not form a complete Cartesian grid; Native regular grid deliberately rejects missing cells.
- Logarithmic colour normalization requires positive Z values. For signed data, use linear, symmetric-log, or diverging-around-center normalization.
- Ensure custom contour levels contain at least two distinct finite numbers.

A plot style or LaTeX fails

- Reinstall requirements to provide SciencePlots.
- Disable Use LaTeX text if a LaTeX distribution is not installed.
- Remove unsupported tokens from Custom styles.

Current scope limitations

The current version does not provide:

- Excel export, legacy .xls import, named/range-only Excel import, database, JSON/XML, HDF5, NetCDF, MATLAB, or instrument-format connectors.
- Encoding selection, arbitrary non-leading row filtering during import, or different forced parser overrides for individual files inside one batch.
- A workbook/folder hierarchy, matrix sheets, rich cell formatting, notes, or dependency/recalculation graphs.
- Multi-condition filters, pivot tables, resampling, date/time tooling, or reusable preparation recipes.
- Multi-panel pages, true 3D plots, polar/ternary plots, annotation objects, or general graph templates.
- Global multi-dataset fitting, automated model ranking, peak deconvolution, or surface fitting.
-

A general scripting console, plugins, autosave/recovery, or installer. Restricted Python API recordings and batch recipes are available as described above. Disk-backed large-table browsing is available, but full-source analysis controls, direct large-table editing, and vendor-map disk storage are not yet implemented.

Consult Roadmap.md for planned work; roadmap entries are not instructions for currently available features.

---

16. Quick Reference

Recommended end-to-end workflow

1. Import/create data.
2. Assign column names, roles, types, units, and long names.
3. Clean data with non-destructive Data Preparation outputs.
4. Apply formulas or analysis operations, using preflight for batch work.
5. Inspect generated comments and batch summaries.
6. Create a plot document, configure axes/legend/locators, and save the project.
7. Export final data/figures as needed.
8. Save the .jasc project to retain the editable workspace.

Example data by feature family

Start with Test_Data/README.txt, then Test_Data/Feature_Examples/README.txt. Each numbered directory contains ready-to-use synthetic inputs, expected.json with independent reference answers, and a GUI walkthrough. Thirteen families also include run_example.py for restricted replay. Start an empty project, select Macros > Run Python API Script, and use the numbered family directory as the allowed input folder, not its input subfolder. Select a separate output parent; each run creates a fresh results folder. The runner retains the completed workspace as _jasc_result.jasc. Inputs do not need to be regenerated before use.

Directory under Test_Data/Feature_Examples	Features and expected behavior
01_Import	Text delimiters/headers/preambles, encodings, decimal comma and two Excel tabs; numeric controls follow y=2x+1.
02_Worksheet_Editing	Smart Fill, paste, Fill Range, metadata, undo/redo and sheet lifecycle; numeric seeds fill to x=1..6 and y=10..60.
03_Data_Preparation	Missing values, duplicates, sort, filter, subset, group, join, bin and melt; ten raw rows become seven clean rows.
04_Formulas	Safe expressions, cross-sheet background subtraction and selected rows; corrected=2.5x+4.
05_Signal_Processing	Six smoothing methods, calculus, FFT/iFFT; known peaks at 5 and 13 Hz and inverse reconstruction.
06_Interpolation	1D, 2D and N-D prediction with known affine answers and explicit outside-range blanks.
07_Statistics	Single/pair/multi/group complete reports, PCA and companion data; three groups of twenty and a regression control.

08_Curve_Fitting	Linear, Gaussian, decay and custom fits with bounds/reports; further model-family responses for GUI exploration.
09_Asymmetric_Uncertainty	Separate percentile endpoints, asymmetric plots and median-likelihood fitting; expected slope about 1.7.
10_Spectral_Analysis	Baseline, detection/review, composite Gaussian fit and six-pixel Raman map with known parameter fields.
11_Plots_and_Maps	Every registered plot subtype, dual Y, map products and comparisons; manual mouse/pop-out/theme checks.
12_Image_Digitization	Manual linear/log XY and colour-map extraction with raster images and analytic reference values; no replay adapter.
13_Projects_and_Export	Ordinary project round-trip and CSV/TSV/PNG/SVG/PDF export; encryption/signing remain manual.
14_Large_Data	Disk-backed source and explicit 400-row extraction from a 12,000-row fixture, with window provenance.

Test_Data/Macro_Example/README.txt provides the additional whole-folder macro workflow described in Chapter 13. Existing surface-map and legacy project samples remain available. These are representative feature-family demonstrations, not an exhaustive validation of every model, optimizer, instrument dialect or statistical assumption. Raster review, plot mouse interactions, theming and protected project operations have manual steps. The large-data fixture demonstrates paging semantics, not larger-than-RAM performance. Developer regeneration/testing commands are in the catalog guide; they do not rebuild the executable.

Output type summary

Operation	Primary output
Fill Range / Formula / Derivative / Integral / Smoothing	Column(s) written into target worksheet(s)
Data Preparation	New non-destructive dataset
FFT / Inverse FFT / Interpolation	New result dataset(s)
Statistics	New report/result dataset(s), optional plot
Curve Fit	Fit dataset, report, and optional residual/covariance/plot outputs
Figure digitizer	New calibrated dataset with diagnostics
Plot Inspector	Editable plot draft; optionally saved plot definition

Naming tokens

Token	Expands to	Common locations

{source}	Source worksheet name	FFT/iFFT, interpolation, statistics, fit result sheets
{column}	Processed Y/signal/value column	Formula/math/smoothing columns and result sheets where supported

Metadata role hints

Role	Typical use
X	Independent variable, time, horizontal coordinate
Y	Dependent signal/value
Z	Surface/map value
Error	Generic uncertainty fallback
X error / Y error	Errorbar uncertainty and explicit fit-uncertainty choices
X/Y lower bound / upper bound	Endpoint columns; percentile levels and centre/origin are chosen explicitly in plot/fit controls
Group	Statistics factor/group default
Label	Categorical/composition labels
Weight	Semantic hint for an explicitly selected curve-fit weight

Safe practices

- Save the project before opening or closing another workspace and before exiting JASC.
- Prefer unique-output policies until a batch preflight has been reviewed.
- Assign explicit Numeric/Text types when automatic inference could be ambiguous.
- Use exact-name batch binding for standardized headers and position binding only when layouts are known to match.
- Leave Live preview off while editing expensive 2D-map settings on large datasets; use Update Series and Refresh Plot deliberately.
- Treat exported CSV/TXT and figures as deliverables; treat .jasc as the editable project source.